

Random Variables: Probability Distribution and Expectation

2.1 RANDOM VARIABLE

Definition 1: Let S be the sample space (or event space) associated with a given random experiment. Then a real valued function X defined on S is called a **one dimensional random variable** or just a **random variable (r.v.)** or sometimes a **variate**.

In other words, a **random variable** is defined as a variable which takes numerical values determined by the outcomes of a random experiment. So, a random variable can be thought of as a function that maps the points of the sample space into the set of real numbers.

The sample space S is termed as the **domain** of the corresponding random variable and the collection of all the numbers (values of random variable) is termed as the **range** or **spectrum** of the random variable.

Notes: (i) One dimensional random variables will be denoted by capital letters X, Y, Z , etc.

(ii) Let w be an outcome of the underlying random experiment, then $X(w)$ represents the real number which the random variable X associates with the outcome w . The values taken by a random variable X are usually denoted by lower case letters x, y , etc.

(iii) Two or more different outcomes might give the same value of X but two different numbers in the range (or spectrum) cannot be assigned to the same outcome.

Example 1: Consider the random experiment of tossing a coin twice. Here the sample space is $S = \{HH, HT, TH, TT\}$.

Let X represents the random variable associated with the outcome 'number of heads'. Then we can assign a number for X as shown in the table below:

Outcome	HH	HT	TH	TT
X	2	1	1	0

Example 2: Consider the random experiment of throwing a die. Here the sample space is $S = \{1, 2, 3, 4, 5, 6\}$.

The most natural choice of random variable X is $X(w) = w$, where $w = 1, 2, 3, 4, 5, 6$. Another random variable Y on S can be defined as:

$$Y(w) = \begin{cases} 1, & \text{if } w \text{ is even} \\ 0, & \text{if } w \text{ is odd} \end{cases}$$

So, when the face turned up is 2 or 4 or 6, Y takes the value 1 and when 1 or 3 or 5 come up, Y takes the value 0.

Note: Associated with same sample space different random variables can be defined.

Definition 2 (Events described by a random variable): If X is a random variable and x is a fixed real number, then the event $(X = x)$ is defined as:

$$(X = x) = \{w \in S : X(w) = x\}$$

Similarly, we can define the following events:

$$(X \leq a) = \{w \in S : X(w) \leq a\}$$

$$(X > b) = \{w \in S : X(w) > b\}$$

$$(a < X \leq b) = \{w \in S : a < X(w) \leq b\}$$

The corresponding probabilities are:

$$P(X = x) = P\{w \in S : X(w) = x\}$$

$$P(X \leq a) = P\{w \in S : X(w) \leq a\}$$

$$P(X > b) = P\{w \in S : X(w) > b\}$$

$$P(a < X \leq b) = P\{w \in S : a < X(w) \leq b\}.$$

Notes: (i) The symbol ' $\in$ ' stands for the word 'belongs to'. (ii) The symbol ':' stands for the word 'such that'.

Example 3: Consider the random experiment given in Example 1.

Let A and B denote the events $(X = 2)$ and $(X \leq 1)$ respectively. Here the sample space is $S = \{HH, HT, TH, TT\}$ and X represents the r.v. associated with the outcome 'number of heads'.

$$\therefore A = (X = 2) = \{w \in S : X(w) = 2\} = \{HH\}$$

and
$$B = (X \leq 1) = \{w \in S : X(w) \leq 1\} = \{HT, TH, TT\}.$$

Assume that the event points (or sample points or outcomes) are equally likely, we have

$$P(A) = P(X = 2) = \frac{1}{4} \text{ and } P(B) = P(X \leq 1) = \frac{3}{4}.$$

Theorem 1: If X_1 and X_2 are two random variables defined on the same sample space (or event space) S , then $X_1 + X_2$ is also a random variable defined on S .

Proof: Let X_1 and X_2 be two random variables defined on the sample space S associated with a given random experiment E . Let $w \in S$ be an outcome of E , so $X_1(w)$ and $X_2(w)$ are real numbers.

Hence $X_1(w) + X_2(w)$ is also a real number, i.e., $(X_1 + X_2)(w)$ is also a real number. Thus $X_1 + X_2$ is a function from the sample space S to the set of real numbers. Therefore $X_1 + X_2$ is also a random variable on S .

Theorem 2: If X_1 and X_2 are two random variables defined on the same sample space S and C_1, C_2 are constants, then $CX_1, X_1 X_2$ and $C_1 X_1 + C_2 X_2$ are also random variables defined on S .

Proof: Proceed as in Theorem 1.

Note: It follows that $X_1 - X_2$ is also a r.v. on S .

Theorem 3: If X_1 and X_2 are two random variables defined on the same sample space S , then $\max(X_1, X_2)$ is also a random variable on S .

Proof: Proceed as in Theorem 1.

Example 4: A random experiment consists of three independent tosses of a fair (unbiased) coin. The sample space S contains $2^3 = 8$ event points:

$$S = \{HHH, THH, HTH, HHT, HTT, THT, TTH, TTT\}.$$

Let us define two random variables X and Y on S , where X is associated with the outcome 'number of heads' and Y is associated with the outcome 'number of tails'.

Let X takes the values 3, 2, 1, 0 corresponding to three heads, two heads, one head, no head and Y takes the values 3, 2, 1, 0 corresponding to three tails, two tails, one tail, no tail. Then the random variable $X + Y$ takes the value 3 which is shown below:

Outcome	Value of X	Value of Y	Value of $X + Y$
HHH	3	0	3
THH	2	1	3
HTH	2	1	3
HHT	2	1	3
HTT	1	2	3
THT	1	2	3
TTH	1	2	3
TTT	0	3	3

2.2 DISTRIBUTION FUNCTION OF A RANDOM VARIABLE

Most of the information about a random experiment described by a random variable X is obtained by studying the behaviour of a function known as the **distribution function** which is defined as follows:

Definition 3 (Distribution function): Let X be a random variable defined on the sample space S associated with a given random experiment. The **cumulative distribution function (c.d.f.)**, or simply, **distribution function (d.f.)** of X is denoted and defined as

$$F_X(x) = P(X \leq x), -\infty < x < \infty$$

where $P(X \leq x) = \{w \in S : X(w) \leq x\}$.

Sometimes $F_X(x)$ is simply written as $F(x)$.

Clearly, $0 \leq F(x) \leq 1$.

Properties of distribution function: Let $F(x)$ is the distribution function of a random variable X and $a < b$ where a, b are any two real numbers.

$\therefore$

$$F(a) = P(X \leq a) \quad \text{and} \quad F(b) = P(X \leq b)$$



Property 1: (i) $P(a < X \leq b) = F(b) - F(a)$.

Proof: The events $(X \leq a)$ and $(a < X \leq b)$ are mutually exclusive and

$$\begin{aligned} (X \leq b) &= (X \leq a) \cup (a < X \leq b) \\ \Rightarrow P(X \leq b) &= P\{(X \leq a) \cup (a < X \leq b)\} \\ \Rightarrow P(X \leq b) &= P(X \leq a) + P(a < X \leq b) \quad [\text{By addition law}] \\ \Rightarrow P(a < X \leq b) &= P(X \leq b) - P(X \leq a) \\ &= F(b) - F(a) \end{aligned}$$

(ii) $P(a \leq X \leq b) = P(X = a) + F(b) - F(a)$.

Proof: The events $(X = a)$ and $(a < X \leq b)$ are mutually exclusive and

$$\begin{aligned} (a \leq X \leq b) &= (X = a) \cup (a < X \leq b) \\ \Rightarrow P(a \leq X \leq b) &= P\{(X = a) \cup (a < X \leq b)\} \\ \Rightarrow P(a \leq X \leq b) &= P(X = a) + P(a < X \leq b) \quad [\text{By addition law}] \\ \Rightarrow P(a \leq X \leq b) &= P(X = a) + F(b) - F(a) \quad [\text{By (i)}] \end{aligned}$$

(iii) $P(a < X < b) = F(b) - F(a) - P(X = b)$.

Proof: The events $(a < X < b)$ and $(X = b)$ are mutually exclusive and

$$\begin{aligned} (a < X \leq b) &= (a < X < b) \cup (X = b) \\ \Rightarrow P(a < X \leq b) &= P\{(a < X < b) \cup (X = b)\} \\ \Rightarrow F(b) - F(a) &= P(a < X < b) + P(X = b) \quad [\text{By (i) and addition law}] \\ \Rightarrow P(a < X < b) &= F(b) - F(a) - P(X = b). \\ \text{(iv)} \quad P(a \leq X < b) &= P(X = a) + F(b) - F(a) - P(X = b). \end{aligned}$$

Proof: The events $(X = a)$ and $(a < X < b)$ are mutually exclusive and

$$\begin{aligned} (a \leq X < b) &= (X = a) \cup (a < X < b) \\ \Rightarrow P(a \leq X < b) &= P\{(X = a) \cup (a < X < b)\} \\ \Rightarrow P(a \leq X < b) &= P(X = a) + P(a < X < b) \quad [\text{By addition law}] \\ \Rightarrow P(a \leq X < b) &= P(X = a) + F(b) - F(a) - P(X = b). \quad [\text{By (iii)}] \end{aligned}$$

Note: When $P(X = a) = P(X = b) = 0$, the probability of all the four events $(a < X < b)$, $(a \leq X < b)$, $(a < X \leq b)$ and $(a \leq X \leq b)$ is same and is equal to $F(b) - F(a)$.

Property 2: (i) $0 \leq F(x) \leq 1$.

(ii) $x < y \Rightarrow F(x) \leq F(y)$. [W.B.U.T. 2010]

Proof: (i) $F(x) = P(X \leq x)$.

Now, $0 \leq P(X \leq x) \leq 1$.

$\therefore 0 \leq F(x) \leq 1$.

(ii) $x < y \Rightarrow F(y) - F(x) = P(x < X \leq y)$ [By property 1, (i)]

$\Rightarrow F(y) - F(x) \geq 0$ [$\because P(x < X \leq y) \geq 0$]

$\Rightarrow F(x) \leq F(y)$.

$\Rightarrow$ Distribution function is monotonic non-decreasing everywhere.

Property 3: $F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0$ and $F(\infty) = \lim_{x \rightarrow \infty} F(x) = 1$. (W.B.U.T. 2010)

Proof: By definition of distribution function,

$$F(\infty) = \lim_{x \rightarrow \infty} F(x) = \lim_{x \rightarrow \infty} P(X \leq x) = P(S) = 1,$$

$$F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = \lim_{x \rightarrow -\infty} P(X \leq x) = P(\varnothing) = 0.$$

Property 4: The distribution function $F(x)$ is continuous on the right, i.e.,

$$\lim_{h \rightarrow 0+} F(a + h) = F(a). \quad (\text{W.B.U.T. 2010})$$

Proof: By property 1 (ii):

$$P(a \leq X \leq a + h) = P(X = a) + F(a + h) - F(a)$$

$$\therefore \lim_{h \rightarrow 0+} P(a \leq X \leq a + h) = P(X = a) + \lim_{h \rightarrow 0+} F(a + h) - F(a).$$

$$\Rightarrow P(X = a) = P(X = a) + \lim_{h \rightarrow 0+} F(a + h) - F(a).$$

$$\Rightarrow \lim_{h \rightarrow 0+} F(a + h) = F(a).$$

$\Rightarrow F(x)$ is continuous on the right.

Property 5: $P(X = a) = F(a) - \lim_{h \rightarrow 0+} F(a - h) = F(a) - \lim_{h \rightarrow a-} F(x)$
 $= F(a) - F(a - 0).$

Proof: For $h > 0$, using property 1 (i):

$$P(a - h < X \leq a) = F(a) - F(a - h)$$

$$\therefore P(X = a) = \lim_{h \rightarrow 0+} P(a - h < X \leq a) = \lim_{h \rightarrow 0+} \{F(a) - F(a - h)\}$$

$$= F(a) - \lim_{h \rightarrow 0+} F(a - h)$$

$$= F(a) - \lim_{x \rightarrow a-} F(x) = F(a) - F(a - 0).$$

2.3 DISCRETE RANDOM VARIABLES

There are two types of random variables known as:

- (i) discrete random variables,
- (ii) continuous random variables.

Definition 4 (Discrete random variable): If a random variable takes finite or an infinite sequence (countably infinite) of distinct values it is called a **discrete random variable**.

In other words, a real valued function defined on a discrete sample space is known as a **discrete random variable**.

The **range** or **spectrum** of a discrete random variable is **finite** or **countably infinite**.

Example: The random variable X defined in Example 1 and the random variables X, Y defined in Example 2 are discrete random variables.

Definition 5 [Probability mass function (p.m.f.)]: Let X be a discrete random variable which assumes the values $x_0, x_1, x_2, \dots, x_n, \dots$ with probabilities $P(X = x_i) = p_i$. The value of p_i depends on x_i , i.e., on i . This function p_i is called **probability mass function (p.m.f.)** of the random variable X provided the following conditions are satisfied:

$$(i) p_i \geq 0, \forall i \quad (ii) \sum_{i=0}^{\infty} p_i = 1.$$

A particular value of p_i is called a **probability mass** and the set of ordered pairs (x_i, p_i) is known as the **discrete probability distribution** of the random variable X .

Definition 6 (Discrete distribution function): Let X be a discrete random variable assuming the values $x_0, x_1, x_2, \dots, x_n, \dots$, then the **distribution function (d.f.)** of X is given by

$$F(x) = P(X \leq x) = P(X = x_0) + P(X = x_1) + \dots + P(X = x_i) \\ = \sum_{k=0}^i p_k, \text{ where } x_i \leq x < x_{i+1}$$

Example 1: A random variable X has the following probability mass function:

X	0	1	2	3	4
$P(X = x)$	0	$5k$	$3k$	k	k

(W.B.U.T. 2011)

Determine the value of k .

Solution: We know that $P(X = x)$ is a possible probability mass function if $P(X = x) \geq 0, \forall x$ and

$$\sum_x P(X = x) = 1.$$

Here,

$$\sum_x P(X = x) = 1$$

$\Rightarrow$

$$0 + 5k + 3k + k + k = 1$$

$\Rightarrow$

$$10k = 1$$

$\Rightarrow$

$$k = \frac{1}{10}$$

Also, for $k = \frac{1}{10}$, $P(X = x) \geq 0, \forall x$.

Hence the required value of k is $\frac{1}{10}$.

Example 2: A random experiment consists of three independent tosses of an unbiased coin.

Let X and Y denote the following events:

$X \equiv$ The number of heads

$Y \equiv$ Consecutive occurrence of at least two heads.

Find the probability mass function (or probability distribution) of (i) X (ii) Y (iii) $X + Y$ (iv) XY .

Solution: The sample space contains $2^3 = 8$ event points. Let us assign a number to each of X and Y as shown in the following table:

Events	X	Y	X + Y	X Y
HHH	3	1	4	3
THH	2	1	3	2
HTH	2	0	2	0
HHT	2	1	3	2
HTT	1	0	1	0
THT	1	0	1	0
TTH	1	0	1	0
TTT	0	0	0	0

(i) X is a random variable which assumes the values 0, 1, 2, 3.

Values of X					
(x)	0	1	2	3	$p(x) > 0,$
$p(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	$\sum p(x) = 1.$

(ii) Y is a random variable which assumes the values 0, 1.

Values of Y			
(y)	0	1	$p(y) > 0,$
$p(y)$	$\frac{5}{8}$	$\frac{3}{8}$	$\sum p(y) = 1.$

(iii) $U = X + Y$ is a random variable which can take the values 0, 1, 2, 3, 4.

Values of U						
(u)	0	1	2	3	4	$p(u) > 0,$
$p(u)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$\sum p(u) = 1.$

(iv) $V = XY$ is a random variable which can take the values 0, 2, 3.

Values of V				
(v)	0	2	3	$p(v) > 0,$
$p(v)$	$\frac{5}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$\sum p(v) = 1.$

Example 3: A discrete random variable X has the following probability mass function:

Values of X, x	0	1	2	3	4	5	6	7
$p(x)$	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

(i) Find k.

(W.B.U.T. 2011)

(ii) Evaluate $P(X < 6)$, $P(X \geq 6)$ and $P(0 < X < 5)$.

(W.B.U.T. 2004, 2006, 2007)

(iii) Determine the distribution function of X.

(W.B.U.T. 2004)

Solution: (i) Since $\sum_{x=0}^7 p(x) = 1$, we have

$$10k^2 + 9k = 1$$

$$\Rightarrow 10k^2 + 9k - 1 = 0$$

$$\Rightarrow 10k^2 + 10k - k - 1 = 0$$

$$\Rightarrow (10k - 1)(k + 1) = 0$$

$$\Rightarrow k = \frac{1}{10} \quad (\because k \neq -1, \text{ as } p(x) \geq 0, \forall x)$$

$$(ii) P(X < 6) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5)$$

$$= 0 + \frac{1}{10} + \frac{2}{10} + \frac{2}{10} + \frac{3}{10} + \frac{1}{100} \quad \left(\because k = \frac{1}{10} \right)$$

$$= \frac{81}{100}$$

$$P(X \geq 6) = 1 - P(X < 6) = 1 - \frac{81}{100} = \frac{19}{100}$$

$$P(0 < X < 5) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

$$= 8k = \frac{4}{5}$$

(iii) The distribution function $F(x)$ of X is given below:

$$F(x) = \begin{cases} 0 & , -\infty < x < 1 \\ \frac{1}{10} & , 1 \leq x < 2 \\ \frac{1}{10} + \frac{2}{10} = \frac{3}{10} & , 2 \leq x < 3 \\ \frac{3}{10} + \frac{2}{10} = \frac{1}{2} & , 3 \leq x < 4 \\ \frac{1}{2} + \frac{3}{10} = \frac{8}{10} & , 4 \leq x < 5 \\ \frac{8}{10} + \left(\frac{1}{10}\right)^2 = \frac{81}{100} & , 5 \leq x < 6 \\ \frac{81}{100} + 2\left(\frac{1}{10}\right)^2 = \frac{83}{100} & , 6 \leq x < 7 \\ \frac{83}{100} + 7\left(\frac{1}{10}\right)^2 + \frac{1}{10} = 1 & , x \geq 7. \end{cases}$$

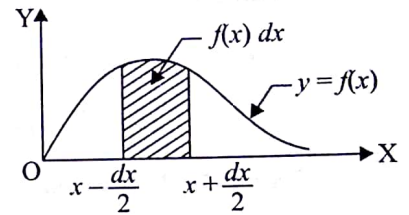
Note: If $P(X \leq a) > \frac{1}{2}$, then the minimum value of a is 4.

(W.B.U.T. 2004)

2.4 CONTINUOUS RANDOM VARIABLES

Definition 7 (Continuous random variable): A random variable X is said to be continuous if it can assume all possible values between certain limits, i.e., it can take all possible values in a given interval. In other words, a random variable is said to be continuous when its different values cannot be put in one to one correspondence with a set of integers.

Probability density function: Consider the small interval $\left[x - \frac{dx}{2}, x + \frac{dx}{2}\right]$ of x . Let $y = f(x)$ be any continuous function of x such that $f(x) dx$ represents the probability that X falls in the infinitesimal interval $\left[x - \frac{dx}{2}, x + \frac{dx}{2}\right]$, symbolically, $P\left(x - \frac{dx}{2} \leq X \leq x + \frac{dx}{2}\right) = f(x) dx$. Now, $f(x) dx$ represents the area bounded by the curve $y = f(x)$, x -axis and the ordinates at the points $x - \frac{dx}{2}$, $x + \frac{dx}{2}$ as shown in the adjacent diagram.



Definition 8 (Probability density function): If X is a continuous random variable such that $P\left(x - \frac{dx}{2} \leq X \leq x + \frac{dx}{2}\right) = f(x) dx$, then $f(x)$ is called the **probability density function (p.d.f.)** of X provided $f(x)$ satisfies the following conditions:

$$(i) f(x) \geq 0, \forall x \in R$$

$$(ii) \int_R f(x) dx = 1$$

where R is the collection of all points in the entire range or spectrum of the random variable X .

$$(iii) P(E) = \int_E f(x) dx$$

where E is any well defined event.

Notes: (i) The probability for a variate value to lie in the interval of length dx is $f(x) dx$ and hence the probability for a variate value to fall in the finite interval $[a, b]$ is

$$P(a \leq X \leq b) = \int_a^b f(x) dx,$$

which represents the area between the curve $y = f(x)$, x -axis and the ordinates at $x = a$ and $x = b$. Further, since total probability is unity, we have $\int_a^b f(x) dx = 1$, where $[a, b]$ is the range of the random variable X , i.e., $a \leq X \leq b$.

$$(ii) P(X = a) = P(a \leq X \leq a) = \int_a^a f(x) dx = 0.$$

Therefore, probability at a particular point is zero, i.e., it is impossible that a continuous random variable assumes a specific value.

(iii) Let $F(x)$ be the distribution function of a continuous random variable X .

Using Property 1 of Art. 2.2 and (i), (ii), we get

$$P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b) = \int_a^b f(x) dx = F(b) - F(a).$$

$\Rightarrow f(x) dx = dF(x)$. This is known as **probability differential** of X .

$\Rightarrow f(x) = F'(x)$.

(iv) **Density curve:** $y = f(x)$ is known as the **probability density curve** or **probability curve**. It gives the graphical representation of the corresponding continuous distribution.

Definition 9 (Continuous distribution function): If X is a continuous random variable with probability density function $f(x)$, then the function

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt, \quad -\infty < x < \infty$$

is called the **distribution function (d.f.)** of X .

Notes: (i) $0 \leq F(x) \leq 1, -\infty < x < \infty$.

$$(ii) F(-\infty) = \lim_{x \rightarrow -\infty} \int_{-\infty}^x f(t) dt = 0$$

$$\text{and } F(\infty) = \lim_{x \rightarrow \infty} \int_{-\infty}^x f(t) dt = \int_{-\infty}^{\infty} f(t) dt = 1.$$

Example 1: Find the constant k so that the function

$$f(x) = \begin{cases} kx^2, & 0 < x < 3 \\ 0, & \text{otherwise} \end{cases}$$

is a probability density function (p.d.f.). Find the distribution function and evaluate $P(1 < X < 2)$.

Solution: From the property of p.d.f. and given condition, we get

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_0^3 kx^2 dx = 1$$

$$\Rightarrow k \left[\frac{x^3}{3} \right]_0^3 = 1$$

$$\Rightarrow 9k = 1$$

$$\therefore k = \frac{1}{9}.$$

Also, $f(x) \geq 0, \forall x$ and $k = \frac{1}{9}$. So, $f(x)$ is a possible p.d.f. for $k = \frac{1}{9}$.

The distribution function of X is given by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

When $x \leq 0$:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt = 0 \quad (\because f(t) = 0, \forall t \leq 0)$$

When $0 < x < 3$:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^0 0 dt + \int_0^x \frac{t^2}{9} dt$$

$$= \frac{1}{9} \left[\frac{t^3}{3} \right]_0^x = \frac{x^3}{27}.$$

When $x \geq 3$:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^0 0 dt + \int_0^3 \frac{t^2}{9} dt + \int_3^x 0 dt = 1$$

Hence, the distribution function of X is:

$$F(x) = \begin{cases} 0 & , \quad x \leq 0 \\ \frac{x^3}{27} & , \quad 0 < x < 3 \\ 1 & , \quad x \geq 3 \end{cases}$$

Now,

$$\begin{aligned} P(1 < X < 2) &= \int_1^2 f(x) dx = \int_1^2 \frac{x^2}{9} dx = \frac{1}{9} \left[\frac{x^3}{3} \right]_1^2 \\ &= \frac{1}{27} (2^3 - 1^3) = \frac{7}{27}. \end{aligned}$$

Note: $P(1 < X < 2) = F(2) - F(1) = \frac{2^3}{27} - \frac{1^3}{27} = \frac{7}{27}.$

Example 2: The time one has to wait for a bus at a bus stand is observed to be a random phenomenon governed by the r.v. X with the following p.d.f.:

$$\begin{aligned} f(x) &= 0 & : \quad x < 0 \\ &= \frac{1}{9}(x+1) & : \quad 0 \leq x < 1 \\ &= \frac{4}{9}\left(x - \frac{1}{2}\right) & : \quad 1 \leq x < \frac{3}{2} \\ &= \frac{4}{9}\left(\frac{5}{2} - x\right) & : \quad \frac{3}{2} \leq x < 2 \\ &= \frac{1}{9}(4-x) & : \quad 2 \leq x < 3 \\ &= \frac{1}{9} & : \quad 3 \leq x < 6 \\ &= 0 & : \quad x \geq 6. \end{aligned}$$

Let events A and B are defined as follows:

$A \equiv$ One waits between 0 and 2 min. inclusive

$B \equiv$ One waits between 1 and 3 min. inclusive

Show that (i) $P(B/A) = \frac{2}{3}$ (ii) $P(\bar{A}\bar{B}) = \frac{1}{3}.$

Solution: (i) By definition, $P(B/A) = \frac{P(AB)}{P(A)}$... (1)

Now,

$$P(A) = \int_0^2 f(x) dx = \int_0^1 \frac{1}{9}(x+1) dx + \int_1^{3/2} \frac{4}{9}\left(x - \frac{1}{2}\right) dx + \int_{3/2}^2 \frac{4}{9}\left(\frac{5}{2} - x\right) dx$$

$$= \frac{1}{9} \left[\frac{x^2}{2} + x \right]_0^1 + \frac{4}{9} \left[\frac{x^2}{2} - \frac{x}{2} \right]_1^{3/2} + \frac{4}{9} \left[\frac{5}{2}x - \frac{x^2}{2} \right]_{3/2}^2 = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2},$$

$$\begin{aligned} \text{Again, } P(AB) &= P(1 \leq X \leq 2) = \int_1^2 f(x) dx = \int_1^{3/2} \frac{4}{9} \left(x - \frac{1}{2} \right) dx + \int_{3/2}^2 \frac{4}{9} \left(\frac{5}{2} - x \right) dx \\ &= \frac{1}{6} + \frac{1}{6} = \frac{1}{3}. \end{aligned}$$

Therefore, from (1), we get

$$P(B/A) = \frac{1/3}{1/2} = \frac{2}{3}.$$

(ii) $\bar{A}\bar{B}$ $\equiv$ Waiting time is more than 3 min.

$$\begin{aligned} \therefore P(\bar{A}\bar{B}) &= P(X > 3) = \int_3^{\infty} f(x) dx = \int_3^6 \frac{1}{9} dx + \int_6^{\infty} 0 dx \\ &= \frac{1}{9} [x]_3^6 = \frac{3}{9} = \frac{1}{3}. \end{aligned}$$

Example 3: The probability density function is given by $f(x) = kx^2$, $0 \leq x \leq 6$; $f(x) = k(12-x)^2$, $6 \leq x \leq 12$; $f(x) = 0$, elsewhere.

(W.B.U.T. 2007)

(i) Evaluate the constant k . (ii) Find $P(6 \leq X \leq 9)$.

Solution: (i) The given function $f(x)$ is a possible probability density function if $f(x) \geq 0$, $\forall x$ and

$$\int_{-\infty}^{\infty} f(x) dx = 1.$$

$$\text{Now, } \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_{-\infty}^0 0 dx + \int_0^6 k x^2 dx + \int_6^{12} k (12-x)^2 dx + \int_{12}^{\infty} 0 dx = 1$$

$$\Rightarrow k \left[\frac{x^3}{3} \right]_0^6 + k \left[-\frac{1}{3} (12-x)^3 \right]_6^{12} = 1$$

$$\Rightarrow 72k + 72k = 1$$

$$\Rightarrow k = \frac{1}{144}$$

$$\therefore f(x) = \begin{cases} \frac{x^2}{144} & , \text{ for } 0 \leq x \leq 6 \\ \frac{(12-x)^2}{144} & , \text{ for } 6 \leq x \leq 12 \\ 0 & , \text{ elsewhere} \end{cases}$$

Obviously, $f(x) \geq 0$, $\forall x$. So, $f(x)$ is a possible p.d.f. for $k = \frac{1}{144}$.

$$\begin{aligned}
 (ii) \quad P(6 \leq X \leq 9) &= \int_6^9 f(x) dx = \int_6^9 \frac{(12-x)^2}{144} dx = \frac{1}{144} \left[-\frac{1}{3}(12-x)^3 \right]_6^9 \\
 &= \frac{1}{144}(-9+72) = \frac{63}{144} = \frac{7}{16}.
 \end{aligned}$$

2.5 MEAN OR EXPECTATION OF A RANDOM VARIABLE

Definition 10 (Mean or expectation): (i) If X is a discrete random variable which can assume values $x_1, x_2, \dots, x_n, \dots$ with respective probability $P(X = x_i) = p_i; i = 1, 2, \dots$, then its **mean or expectation** is denoted and defined as

$$m = E(X) = \sum_{i=1}^{\infty} p_i x_i, \text{ such that } \sum_{i=1}^{\infty} p_i = 1.$$

(ii) If X is a continuous random variable with probability density function $f(x)$ then its **mean or expectation** is denoted and defined as

$$m = E(X) = \int_{-\infty}^{\infty} xf(x) dx, \text{ such that } \int_{-\infty}^{\infty} f(x) dx = 1.$$

We state below (without proof) some properties of mean.

Properties of mean (or expectation)

Let X, Y, Z are random variables and a, b are some real constants.

1. (i) If $X = a$, then $E(X) = a$
 (ii) If $a \leq X \leq b$, then $a \leq E(X) \leq b$.
2. Transformation property:
 (i) If $Y = aX$, then $E(Y) = a E(X)$.
 (ii) If $Y = a \pm bX$, then $E(Y) = a \pm b E(X)$.
3. If $Z = aX \pm bY$, then $E(Z) = a E(X) \pm b E(Y)$.
4. $E(XY) = E(X) E(Y)$, provided X and Y are independent.
5. If $g(x)$ is a continuous function, then

$$\begin{aligned}
 E\{g(X)\} &= \sum p_i g(x_i), \text{ for a discrete distribution} \\
 &= \int_{-\infty}^{\infty} g(x) f(x) dx, \text{ for a continuous distribution}
 \end{aligned}$$

Note: Mean of a random variable X is a measure of location of the density of X .

Example 1: Let X be a discrete random variable assuming values $1, 2, 3, \dots$ and suppose that $E(X)$ exists. Show that $E(X) = \sum_{i=1}^{\infty} P(X \geq i)$.

Solution: Here,

$$\begin{aligned}
 E(X) &= 1 \cdot P(X=1) + 2 \cdot P(X=2) + 3 \cdot P(X=3) + 4 \cdot P(X=4) + \dots \\
 &= \{P(X=1) + P(X=2) + P(X=3) + P(X=4) + \dots\} \\
 &\quad + \{P(X=2) + P(X=3) + P(X=4) + \dots\} \\
 &\quad + \{P(X=3) + P(X=4) + \dots\} + \dots \\
 &= P(X \geq 1) + P(X \geq 2) + P(X \geq 3) + \dots \\
 &= \sum_{i=1}^{\infty} P(X \geq i).
 \end{aligned}$$

Example 2: Let X be a random variable with the following probability mass function:

x	-3	6	9
$p(x)$	$1/6$	$1/2$	$1/3$

Find: (i) $E(X)$ (ii) $E(X^2)$ (iii) $E(2X + 1)^2$.

Solution:

$$(i) \ E(X) = \sum xp(x) = (-3) \times \frac{1}{6} + 6 \times \frac{1}{2} + 9 \times \frac{1}{3} = \frac{11}{2}.$$

$$(ii) \ E(X^2) = \sum x^2 p(x) = 9 \times \frac{1}{6} + 36 \times \frac{1}{2} + 81 \times \frac{1}{3} = \frac{93}{2}$$

$$(iii) \ E(2X + 1)^2 = E(4X^2 + 4X + 1) = 4E(X^2) + 4E(X) + 1$$

$$= 4 \times \frac{93}{2} + 4 \times \frac{11}{2} + 1$$

[By (i) and (ii)]

$$= 186 + 22 + 1 = 209.$$

Example 3: A continuous random variable X is distributed over the interval $[0, 1]$ with p.d.f. $f(x) = ax^2 + bx$, where a and b are constants. If the mean of X is 0.25, find the values of a and b .

Solution: Since $f(x)$ is a p.d.f., we have

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_0^1 (ax^2 + bx) dx = 1$$

$$\Rightarrow \left[a \frac{x^3}{3} + b \frac{x^2}{2} \right]_0^1 = 1$$

$$\Rightarrow 2a + 3b = 6 \quad \dots(1)$$

Given, $E(X) = 0.25$

$$\Rightarrow \int_{-\infty}^{\infty} xf(x) dx = 0.25$$

$$\Rightarrow \int_0^1 x(ax^2 + bx) dx = \frac{1}{4}$$

$$\Rightarrow \left[\frac{ax^4}{4} + \frac{bx^3}{3} \right]_0^1 = \frac{1}{4}$$

$$\Rightarrow 3a + 4b = 3 \quad \dots(2)$$

Solving (1) and (2), we get

$$a = -15, b = 12.$$

Example 4: A random variable X has the density function $f(x) = x$, $0 \leq x \leq 1$; $f(x) = \frac{1}{2}$, $1 < x \leq 2$. Find the mean of X . (W.B.U.T. 2011)

Solution:

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} xf(x) dx = \int_0^1 x \cdot x dx + \int_1^2 x \cdot \frac{1}{2} dx \\ &= \left[\frac{x^3}{3} \right]_0^1 + \frac{1}{2} \left[\frac{x^2}{2} \right]_1^2 \\ &= \frac{1}{3} + \frac{1}{2} \left(2 - \frac{1}{2} \right) = \frac{1}{3} + \frac{3}{4} = \frac{13}{12}, \end{aligned}$$

this is the required mean of X .

2.6 VARIANCE AND STANDARD DEVIATION

Definition 11 (Variance): (i) If X is a discrete random variable which can assume values $x_1, x_2, \dots, x_n$, ...with respective probability $P(X = x_i) = p(x_i) = p_i$; $i = 1, 2, \dots$, and mean m , then its **variance** is denoted and defined as

$$\text{Var}(X) = E\{(X - m)^2\} = E\{[X - E(X)]^2\} = \sum_{i=1}^{\infty} (x_i - m)^2 p_i.$$

(ii) If X is a continuous random variable with probability density function $f(x)$ and mean m , then its **variance** is denoted and defined as

$$\text{Var}(X) = E\{(X - m)^2\} = E\{[X - E(X)]^2\} = \int_{-\infty}^{\infty} (x - m)^2 f(x) dx.$$

Definition 12 [Standard Deviation (S.D.):] The **standard deviation (S.D.)** of a random variable X is denoted by $\sigma(X)$ or σ_x or simply σ and is defined as the positive square root of $\text{Var}(X)$

$$\therefore \sigma = +\sqrt{\text{Var}(X)} \Rightarrow \sigma^2 = \text{Var}(X).$$

Note: Variance of a random variable X is a measure of the spread or dispersion of the density of X . Also $\text{Var}(X) = 0$ implies that the whole mass is concentrated about the mean.

Properties of variance: Let X is a random variable and a, b are some real constants.

- (i) $\text{Var}(X) = E(X^2) - \{E(X)\}^2 = E(X^2) - m^2$, where $m = E(X)$.
- (ii) $\text{Var}(aX + b) = a^2 \text{Var}(X)$
- (iii) $\text{Var}(a) = 0$
- (iv) $\text{Var}(X) = E\{X(X - 1)\} - m(m - 1)$, where $m = E(X)$.

Proof:

$$\begin{aligned} \text{(i)} \quad \text{L.H.S.} &= \text{Var}(X) = E\{(X - m)^2\} = E(X^2 - 2mX + m^2) \\ &= E(X^2) - E(2mX) + E(m^2) \\ &= E(X^2) - 2mE(X) + m^2 \\ &= E(X^2) - 2m \cdot m + m^2 = E(X^2) - m^2 = \text{R.H.S.} \end{aligned}$$

[see Property 2 (ii), Art. 2.5]

$$\begin{aligned}
 & E(aX + b) = a E(X) + b \\
 \text{(ii) Now,} \quad & \therefore \text{L.H.S.} = \text{Var}(aX + b) = E[\{aX + b - E(aX + b)\}^2] \\
 & = E[a^2\{X - E(X)\}^2] = a^2 E[\{X - E(X)\}^2] \\
 & = a^2 \text{Var}(X) = \text{R.H.S.} \\
 \text{(iii)} \quad & \text{L.H.S.} = \text{Var}(a) = E[\{a - E(a)\}^2] = E[(a - a)^2] = 0 = \text{R.H.S.} \\
 \text{(iv)} \quad & \text{L.H.S.} = \text{Var}(X) = E\{(X - m)^2\} = E(X^2 - 2mX + m^2) \\
 & = E(X^2 - X + X - 2mX + m^2) \\
 & = E(X^2 - X) + E(X) - E(2mX) + E(m^2) \\
 & = E\{X(X - 1)\} + m - 2mE(X) + m^2 \\
 & = E\{X(X - 1)\} + m - 2m \cdot m + m^2 \\
 & = E\{X(X - 1)\} - m(m - 1) \\
 & = \text{R.H.S.}
 \end{aligned}$$

Definition 13 [Coefficient of Variation (C.V.)]: The ratio of the standard deviation to the mean is known as the **coefficient of variation (C.V.)**. Note that this measure is **unit free** and is often expressed as a percentage.

$$\therefore \text{Coefficient of variation} = \frac{\text{S.D.}}{\text{Mean}} \times 100.$$

Example 1: Given the following probability distribution of X :

x	-3	-2	-1	0	1	2	3
$p(x)$	0.05	0.10	0.30	0	0.35	0.10	0.10

Compute (i) $E(X)$ (ii) $E(2X \pm 3)$ (iii) $E(X^2)$ (iv) $\text{Var}(X)$ (v) $\text{Var}(2X \pm 1)$.

Solution: Observe that $\sum p(x) = 1$ and $p(x) \geq 0, \forall x$.

$$\begin{aligned}
 \text{(i)} \quad E(X) &= \sum xp(x) = (-3)(0.05) + (-2)(0.10) + (-1)(0.30) + 0 \\
 &\quad + 1(0.35) + 2(0.10) + 3(0.10) = 0.20 \\
 \text{(ii)} \quad E(2X \pm 3) &= 2E(X) \pm 3 = 2(0.20) \pm 3 = 0.40 \pm 3 \\
 &= 3.40, -2.60 \\
 \text{(iii)} \quad E(X^2) &= \sum x^2p(x) = (-3)^2(0.05) + (-2)^2(0.10) \\
 &\quad + (-1)^2(0.30) + 0 + 1^2(0.35) + 2^2(0.10) + 3^2(0.10) \\
 &= 0.45 + 0.4 + 0.3 + 0.35 + 0.4 + 0.9 \\
 &= 2.8 \\
 \text{(iv)} \quad \text{Var}(X) &= E(X^2) - \{E(X)\}^2 = 2.8 - (0.20)^2 = 2.76. \\
 \text{(v)} \quad \text{Var}(2X \pm 1) &= 2^2 \text{Var}(X) \quad [\because \text{Var}(aX \pm b) = a^2 \text{Var}(X)] \\
 &= 4 \times 2.76 = 11.04.
 \end{aligned}$$

Example 2: A continuous random variable X has the following p.d.f.:

$$f(x) = \begin{cases} \frac{1}{4}(x+2) & , -1 < x < 1 \\ 0 & , \text{elsewhere} \end{cases}$$

Find the mean and variance of X .

Solution: Observe that $\int_{-\infty}^{\infty} f(x) dx = \int_{-1}^1 \frac{1}{4}(x+2) dx = \left[\frac{1}{4} \left(\frac{x^2}{2} + 2x \right) \right]_{-1}^1 = 1.$

Also, $f(x) \geq 0, \forall x.$

So, $f(x)$ is a possible probability density function.

Now,

$$\text{mean} = E(X) = \int_{-\infty}^{\infty} xf(x) dx = \int_{-1}^1 x \cdot \frac{1}{4}(x+2) dx$$

$$= \frac{1}{4} \left[\frac{x^3}{3} + 2 \frac{x^2}{2} \right]_{-1}^1 = \frac{1}{4} \left(\frac{1}{3} + \frac{1}{3} \right) = \frac{1}{6}.$$

$$\text{Var}(X) = \int_{-\infty}^{\infty} \{x - E(X)\}^2 f(x) dx$$

$$= \int_{-1}^1 \left(x - \frac{1}{6} \right)^2 \frac{1}{4}(x+2) dx$$

$$= \frac{1}{4} \int_{-1}^1 \left(x^2 - \frac{x}{3} + \frac{1}{36} \right) (x+2) dx$$

$$= \frac{1}{4} \int_{-1}^1 \left(x^3 + \frac{5x^2}{3} - \frac{23}{36}x + \frac{1}{18} \right) dx$$

$$= \frac{1}{4} \left[\frac{x^4}{4} + \frac{5x^3}{9} - \frac{23}{72}x^2 + \frac{x}{18} \right]_{-1}^1$$

$$= \frac{1}{4} \left(\frac{10}{9} + \frac{2}{18} \right) = \frac{11}{36}.$$

Note:

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) = \int_{-1}^1 x^2 \cdot \frac{1}{4}(x+2) dx = \frac{1}{4} \left[\frac{x^4}{4} + \frac{2}{3}x^3 \right]_{-1}^1.$$

$$= \frac{1}{4} \left(\frac{2}{3} + \frac{2}{3} \right) = \frac{1}{3}.$$

$\therefore$

$$\text{Var}(X) = E(X^2) - \{E(X)\}^2 = \frac{1}{3} - \left(\frac{1}{6} \right)^2 = \frac{11}{36}.$$

MISCELLANEOUS EXAMPLES

Example 1: If

$$F(x) = 0, \quad -\infty < x < 0$$

$$= \frac{1}{5}, \quad 0 \leq x < 1$$

$$= \frac{3}{5}, \quad 1 \leq x < 3$$

$$= 1, \quad 3 \leq x < \infty,$$

show that $F(x)$ is a possible distribution function. Determine the spectrum and probability mass function (p.m.f.) of the random variable X . (W.B.U.T. 2003)

Solution: Clearly $0 \leq F(x) \leq 1$, $F(x)$ is monotonic non-decreasing everywhere, continuous on right at each point and $F(-\infty) = 0$, $F(\infty) = 1$. Hence $F(x)$ is a possible distribution function with spectrum $\{0, 1, 3\}$ and probability mass function of X is given below.

Using Property 5, Art. 2.2:

$$P(X=0) = F(0) - \lim_{x \rightarrow 0-} F(x) = \frac{1}{5} - 0 = \frac{1}{5}$$

$$P(X=1) = F(1) - \lim_{x \rightarrow 1-} F(x) = \frac{3}{5} - \frac{1}{5} = \frac{2}{5}$$

$$P(X=3) = F(3) - \lim_{x \rightarrow 3-} F(x) = 1 - \frac{3}{5} = \frac{2}{5}$$

So, the distribution of the random variable X is

x_i	0	1	3
p_i	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{2}{5}$

Notes: (i) A function $F(x)$ is said to have a **jump discontinuity** at $x = a$ if both the limits $F(a-0)$ and $F(a+0) = \lim_{x \rightarrow a+} F(x)$ exist and are unequal. $F(a+0) - F(a-0)$ is called the **height of jump** at $x = a$.

(ii) Here $F(x)$ is **piecewise constant**. Moreover, $F(x)$ has a jump discontinuity at each of the points $x = 0, 1, 3$; it is continuous on the right but discontinuous on the left. The heights of jumps at $x = 0, 1, 3$ are respectively $1/5, 2/5, 2/5$. Such a function is called a **step function** having steps of heights $1/5, 2/5, 2/5$ at the points $0, 1, 3$ respectively. This definition may be easily generalized.

Example 2: A box contains 3 red and 5 blue balls. Four balls are drawn at random without replacement. Find the probability distribution of the number of red balls drawn.

Solution: The given box contains $(3 + 5) = 8$ balls. Four balls are drawn at random one after another without replacement, so first ball may be drawn in ${}^8C_1 = 8$ ways, second ball in ${}^7C_1 = 7$ ways, third ball in ${}^6C_1 = 6$ ways and fourth ball in ${}^5C_1 = 5$ ways.

$\therefore$ Total number of outcomes is $n = 8 \times 7 \times 6 \times 5 = \frac{8!}{4!} = {}^8P_4$. (Using principle of counting)

Let X be the random variable denoting the number of red balls drawn from the given box. Then the spectrum of X is the set $\{0, 1, 2, 3\}$.

Now,
$$P(X=0) = \frac{{}^5P_4}{{}^8P_4} = \frac{5!}{(5-4)!} \times \frac{(8-4)!}{8!} = \frac{4 \times 3 \times 2}{8 \times 7 \times 6} = \frac{1}{14},$$

$$P(X=1) = \frac{{}^3C_1 \times {}^5C_3 \times 4!}{{}^8P_4} = 3 \times \frac{5!}{3!(5-3)!} \times \frac{4!}{8!} = \frac{3 \times 2 \times 4 \times 3 \times 2}{8 \times 7 \times 6} = \frac{3}{7},$$

$$P(X=2) = \frac{{}^3C_2 \times {}^5C_2 \times 4!}{{}^8P_4} = 3 \times \frac{5!}{2!3!} \times 4! \times \frac{4!}{8!} = \frac{3}{7},$$

$$P(X=3) = \frac{{}^3C_3 \times {}^5C_1 \times 4!}{{}^8P_4} = \frac{5 \times 4! \times 4!}{8!} = \frac{4 \times 3 \times 2}{8 \times 7 \times 6} = \frac{1}{14}$$

Hence, the required distribution of X is

x_i	0	1	2	3
p_i	$\frac{1}{14}$	$\frac{3}{7}$	$\frac{3}{7}$	$\frac{1}{14}$

Note: $p_i \geq 0, \forall i$ and $\sum_i p_i = 1$.

Example 3: Consider the random experiment of tossing an unbiased coin till a head appears for the first time. Let X be the number of tosses required. Find the distribution of X .

Solution: The spectrum of X is the set $\{1, 2, 3, \dots\}$. Let 'H' denotes the outcome 'head in the first toss', 'TH' denotes the outcome 'tail in the first toss and head in the second toss', etc.

$$\therefore P(X=1) = P(H) = \frac{1}{2},$$

$$P(X=2) = P(TH) = P(T) P(H) = \left(\frac{1}{2}\right)^2,$$

$$P(X=3) = P(TTH) = P(T) P(T) P(H) = \left(\frac{1}{2}\right)^3, \dots,$$

$$P(X=n) = P(\underbrace{TT \dots TH}_{(n-1) \text{ times}}) = P(T) P(T) \dots P(T) P(H) = \left(\frac{1}{2}\right)^n$$

and so on, since the events are independent.

Thus the required distribution is

$$P(X=i) = \left(\frac{1}{2}\right)^i, i = 1, 2, 3, \dots$$

Example 4: The distribution function $F(x)$ of a random variable X is defined as follows:

$$\begin{aligned} F(x) &= A, & -\infty < x < -1 \\ &= B, & -1 \leq x < 0 \\ &= C, & 0 \leq x < 2 \\ &= D, & 2 \leq x < \infty \end{aligned}$$

where A, B, C, D are constants. Determine the values of A, B, C, D , it being given that $P(X=0) = \frac{1}{6}$

$$\text{and } P(X > 1) = \frac{2}{3}.$$

(W.B.U.T. 2002, 2004)

Solution: Here $F(x)$ is a distribution function and so,

$$F(\infty) = 1, \text{ i.e., } \lim_{x \rightarrow \infty} F(x) = 1 \Rightarrow D = 1,$$

$$F(-\infty) = 0, \text{ i.e., } \lim_{x \rightarrow -\infty} F(x) = 0 \Rightarrow A = 0.$$

Given, $P(X = 0) = \frac{1}{6}$

$$\Rightarrow F(0) - \lim_{x \rightarrow 0^-} F(x) = \frac{1}{6}$$

$$\Rightarrow C - B = \frac{1}{6} \quad \dots(1)$$

Also, given $P(X > 1) = \frac{2}{3}$

$$\Rightarrow 1 - P(X \leq 1) = \frac{2}{3}$$

$$\Rightarrow 1 - F(1) = \frac{2}{3}$$

$$\Rightarrow 1 - C = \frac{2}{3} \Rightarrow C = \frac{1}{3}.$$

Putting $C = \frac{1}{3}$ in (1), we get

$$B = C - \frac{1}{6} = \frac{1}{3} - \frac{1}{6} = \frac{1}{6}.$$

$$\therefore A = 0, B = \frac{1}{6}, C = \frac{1}{3} \text{ and } D = 1.$$

Example 5: If the random variable X takes the values 1, 2, 3 and 4 such that $2P(X = 1) = 3P(X = 2) = P(X = 3) = 5P(X = 4)$, find the probability distribution of X . (W.B.U.T. 2006)

Solution: Let $P(X = i) = p_i$, where $p_i \geq 0, \forall i = 1, 2, 3, 4$ and $\sum_{i=1}^4 p_i = 1$.

Given, $2P(X = 1) = 3P(X = 2) = P(X = 3) = 5P(X = 4)$

$$\Rightarrow 2p_1 = 3p_2 = p_3 = 5p_4$$

$$\Rightarrow p_1 = \frac{p_3}{2}, p_2 = \frac{p_3}{3} \text{ and } p_4 = \frac{p_3}{5}$$

$$\therefore \sum_{i=1}^4 p_i = 1$$

$$\Rightarrow \frac{p_3}{2} + \frac{p_3}{3} + p_3 + \frac{p_3}{5} = 1.$$

$$\Rightarrow \frac{61}{30} p_3 = 1$$

$$\Rightarrow p_3 = \frac{30}{61}$$

$$\therefore p_1 = \frac{1}{2} p_3 = \frac{15}{61}, p_2 = \frac{1}{3} p_3 = \frac{10}{61}$$

and

$$p_4 = \frac{1}{5} p_3 = \frac{6}{61}$$

Hence the probability distribution of X is

x_i	1	2	3	4
p_i	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{6}{61}$

Example 6: A random variable X has the following probability distribution:

$X = x$	-2	-1	0	1	2	3
$p(x)$	0.1	k	0.2	$2k$	0.3	$3k$

(i) Find k (ii) Evaluate $P(X < 2)$, $P(X \geq 2)$, $P(-2 < X < 2)$ (iii) Find the minimum value of k such that $P(X \leq 2) \geq 0.8$.(iv) Determine the distribution function $F(x)$ of X .

(W.B.U.T. 2005, 2006, 2009)

Solution: (i) We know that $\sum p(x) = 1$

$$\Rightarrow 0.1 + k + 0.2 + 2k + 0.3 + 3k = 1$$

$$\Rightarrow 6k = 1 - 0.6$$

$$\Rightarrow k = \frac{1}{15}$$

Obviously $p(x) \geq 0$, $\forall x$ and $k = \frac{1}{15}$.(ii) Here, $P(X < 2) = P(X = -2) + P(X = -1) + P(X = 0) + P(X = 1)$

$$= 0.1 + k + 0.2 + 2k = 0.3 + \frac{3}{15}$$

$$= 0.5.$$

$$\left(\because k = \frac{1}{15} \right)$$

$$P(X \geq 2) = 1 - P(X < 2) = 1 - 0.5 = 0.5$$

and $P(-2 < X < 2) = P(X = -1) + P(X = 0) + P(X = 1)$

$$= k + 0.2 + 2k = \frac{3}{15} + 0.2 = 0.4.$$

$$\begin{aligned}
 \text{(iii)} \quad P(X \leq 2) &= P(X = -2) + P(X = -1) + P(X = 0) + P(X = 1) + P(X = 2) \\
 &= 0.1 + k + 0.2 + 2k + 0.3 = 0.6 + 3k
 \end{aligned}$$

$$\therefore P(X \leq 2) \geq 0.8 \Rightarrow 0.6 + 3k \geq 0.8 \Rightarrow k \geq \frac{1}{15}.$$

Hence, the minimum value of k is $\frac{1}{15}$.

(iv) The distribution function $F(x)$ of X is given below:

$$F(x) = \begin{cases} 0 & , -\infty < x < -2 \\ 0.1 & , -2 \leq x < -1 \\ 0.1 + \frac{1}{15} = \frac{1}{6} & , -1 \leq x < 0 \\ 0.1 + \frac{1}{15} + 0.2 = \frac{11}{30} & , 0 \leq x < 1 \\ \frac{11}{30} + \frac{2}{15} = \frac{1}{2} & , 1 \leq x < 2 \\ \frac{1}{2} + 0.3 = 0.8 & , 2 \leq x < 3 \\ 0.8 + \frac{3}{15} = 1 & , x \geq 3 \end{cases}$$

Example 7: A random variable X has the following probability mass function:

x	0	1	2	3	4	5	6
$P(X = x)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$

(a) Find the value of k

(b) Find $P(X < 4)$, $P(X \geq 5)$, $P(3 < X \leq 5)$

(c) Obtain the distribution function $F(x)$

(d) What is the smallest value of x for which $P(X \leq x) > 0.5$?

(W.B.U.T. 2011)

Solution: (a) From the properties of probability mass function we know that $\sum P(X = x) = 1$

$$\Rightarrow k + 3k + 5k + 7k + 9k + 11k + 13k = 1$$

$$\Rightarrow 49k = 1 \Rightarrow k = \frac{1}{49}.$$

Obviously, $P(X = x) \geq 0$, $\forall x$ and $k = \frac{1}{49}$.

$$(b) \quad P(X < 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$$

$$= \frac{1}{49} + \frac{3}{49} + \frac{5}{49} + \frac{7}{49} = \frac{16}{49}$$

$$P(X \geq 5) = P(X = 5) + P(X = 6) = \frac{11}{49} + \frac{13}{49} = \frac{24}{49}$$

$$P(3 < X \leq 5) = P(X = 4) + P(X = 5) = \frac{9}{49} + \frac{11}{49} = \frac{20}{49}.$$

(c) The distribution function $F(x)$ of the random variable X is given below:

$$F(x) = \begin{cases} 0 & , -\infty < x < 0 \\ \frac{1}{49} & , 0 \leq x < 1 \\ \frac{4}{49} & , 1 \leq x < 2 \\ \frac{9}{49} & , 2 \leq x < 3 \\ \frac{16}{49} & , 3 \leq x < 4 \\ \frac{25}{49} & , 4 \leq x < 5 \\ \frac{36}{49} & , 5 \leq x < 6 \\ 1 & , x \geq 6 \end{cases}$$

(d) From the distribution function we observe that

$$P(X \leq 3) = F(3) = \frac{16}{49} < 0.5$$

and $P(X \leq 4) = F(4) = \frac{25}{49} > 0.5.$

So, the smallest value of x for which $P(X \leq x) > 0.5$ is 4.

Example 8: The probability mass function of a discrete random variable X is defined as $P(X = 0) = 3k^2$, $P(X = 1) = 4k - 10k^2$, $P(X = 2) = 5k - 1$ and $P(X = x) = 0$ if $x \neq 0, 1, 2$.

(i) Find the value of k

(ii) Find $P(0 < X < 2/X > 0)$

(iii) Obtain the distribution function $F(x)$ of X

(iv) Find the smallest value of x for which $F(x) = P(X \leq x) > \frac{1}{2}$.

Solution: (i) From the properties of probability mass function we know that

$$\begin{aligned} \Rightarrow \quad \Sigma P(X = x) &= 1 \\ \Rightarrow \quad 3k^2 + 4k - 10k^2 + 5k - 1 &= 1 \\ \Rightarrow \quad 7k^2 - 9k + 2 &= 0 \\ \Rightarrow \quad 7k^2 - 7k - 2k + 2 &= 0 \\ \Rightarrow \quad (k - 1)(7k - 2) &= 0 \end{aligned}$$

...(1)

Now, $k = 1 \Rightarrow P(X = 1) = 4 - 10 = -6 < 0$, which is not possible.

$\therefore k \neq 1$. So, from (1), $k = \frac{2}{7}$.

Obviously, $P(X = x) \geq 0, \forall x$ and $k = \frac{2}{7}$.

$$(ii) \quad P(0 < X < 2 / X > 0) = \frac{P\{(0 < X < 2) \cap (X > 0)\}}{P(X > 0)}$$

$$= \frac{P(0 < X < 2)}{P(X > 0)} = \frac{P(X = 1)}{P(X = 1) + P(X = 2)}$$

Now, $P(X = 1) = 4 \cdot \frac{2}{7} - 10 \left(\frac{2}{7}\right)^2 = \frac{16}{49}$

and $P(X = 2) = 5 \cdot \frac{2}{7} - 1 = \frac{3}{7}$

$\therefore P(0 < X < 2 / X > 0) = \frac{16}{37}$.

(iii) The distribution function $F(x)$ of the random variable X is given below:

$$F(x) = P(X \leq x) = \begin{cases} 0 & , -\infty < x < 0 \\ 3k^2 = 3\left(\frac{2}{7}\right)^2 = \frac{12}{49} & , 0 \leq x < 1 \\ \frac{12}{49} + \frac{16}{49} = \frac{28}{49} & , 1 \leq x < 2 \\ 1 & , x \geq 2 \end{cases}$$

(iv) From the distribution function it is obvious that the smallest value of x for which

$$F(x) = P(X \leq x) > \frac{1}{2} \text{ is } x = 1.$$

Example 9: A random variable X has the following p.d.f.:

$$f(x) = \begin{cases} cx^2 & , 0 \leq x \leq 1 \\ 0 & , \text{otherwise} \end{cases}$$

Find: (i) c (ii) $P\left(0 \leq X \leq \frac{1}{2}\right)$.

(W.B.U.T. 2009, 2011)

Solution: (i) The given function $f(x)$ is a possible probability density function (p.d.f.) if $f(x) \geq 0$,

$$\forall x \text{ and } \int_{-\infty}^{\infty} f(x) dx = 1.$$

Now,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$\Rightarrow$

$$\int_{-\infty}^0 0 dx + \int_0^1 cx^2 dx + \int_1^{\infty} 0 dx = 1$$

$$\Rightarrow c \left[\frac{x^3}{3} \right]_0^1 = 1$$

$$\Rightarrow c = 3.$$

$$\therefore f(x) = \begin{cases} 3x^2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Obviously, $f(x) \geq 0, \forall x$. So, $f(x)$ is a possible p.d.f. for $c = 3$.

$$(ii) \quad P\left(0 \leq X \leq \frac{1}{2}\right) = \int_0^{1/2} f(x) dx = \int_0^{1/2} 3x^2 dx = [x^3]_0^{1/2} = \frac{1}{8}$$

Example 10: A continuous random variable X has p.d.f.:

$$f(x) = \begin{cases} 3x^2, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find 'a' and 'b' such that

$$(i) P(X \leq a) = P(X > a)$$

$$(ii) P(X > b) = 0.973.$$

Solution: (i) We know that $P(X \leq a) + P(X > a) = 1$.

Also, by question: $P(X \leq a) = P(X > a)$.

$$\therefore P(X \leq a) = \frac{1}{2}$$

$$\Rightarrow \int_{-\infty}^a f(x) dx = \frac{1}{2}$$

$$\Rightarrow \int_0^a 3x^2 dx = \frac{1}{2}$$

$$\Rightarrow [x^3]_0^a = \frac{1}{2}$$

$$\Rightarrow a^3 = \frac{1}{2}$$

$$\Rightarrow a = 2^{-1/3}.$$

(ii) Given,

$$P(X > b) = 0.973$$

$$\Rightarrow \int_b^{\infty} f(x) dx = 0.973$$

$$\Rightarrow \int_b^1 3x^2 dx = 0.973$$

$$\Rightarrow [x^3]_b^1 = 0.973$$

$$\Rightarrow 1 - b^3 = 0.973$$

$$\Rightarrow b^3 = 1 - 0.973 = 0.027$$

$$\Rightarrow b = 0.3.$$

Example 11: (i) Is the function defined as follows a density function?

$$f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

(ii) If so, determine the probability that the variate having this density will fall in the interval (1, 2)?

(iii) Also find the cumulative probability function $F(2)$?

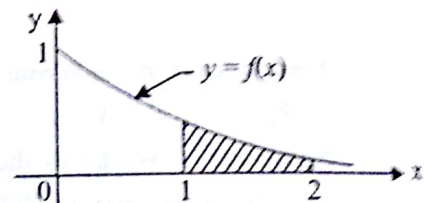
Solution: (i) Here, $\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^0 0 dx + \int_0^{\infty} e^{-x} dx = \lim_{B \rightarrow \infty} \int_0^B e^{-x} dx$
 $= \lim_{B \rightarrow \infty} [-e^{-x}]_0^B = \lim_{B \rightarrow \infty} (1 - e^{-B}) = 1.$

Also, $f(x) \geq 0, \forall x.$

Hence, $f(x)$ is a possible probability density function of a random variable X .

(ii) Required probability $= P(1 < X < 2)$

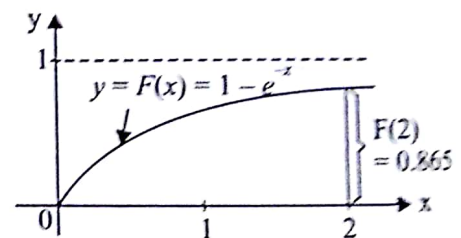
$$\begin{aligned} &= \int_1^2 e^{-x} dx = [-e^{-x}]_1^2 \\ &= e^{-1} - e^{-2} = 0.368 - 0.135 \\ &= 0.233 \end{aligned}$$



This probability is equal to the shaded area as shown in the adjacent figure.

(iii) Cumulative probability function $F(2)$

$$\begin{aligned} &= P(X \leq 2) = \int_{-\infty}^2 f(x) dx = \int_{-\infty}^0 0 dx + \int_0^2 e^{-x} dx \\ &= [-e^{-x}]_0^2 = 1 - e^{-2} = 1 - 0.135 = 0.865 \end{aligned}$$



It is shown in the adjacent figure.

Example 12: Show that the function:

$$f(x) = \begin{cases} \frac{1}{4}, & -2 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

is a possible probability density function. Construct the distribution function and compute

(i) $P(X < 1)$ (ii) $P(|X| > 1)$ (iii) $P(|X - 1| \geq 2)$ (iv) $P(2X + 3 > 5)$

Solution: Observe that $f(x) \geq 0, \forall x$

and $\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{-2} 0 dx + \int_{-2}^2 \frac{1}{4} dx + \int_2^{\infty} 0 dx = \int_{-2}^2 \frac{1}{4} dx = \frac{1}{4} [x]_{-2}^2 = 1.$

Therefore, $f(x)$ is a possible probability density function. The distribution function $F(x)$ is defined by

$$F(x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^x 0 dt = 0, \text{ for } -\infty < x \leq -2$$

$$F(x) = \int_{-\infty}^x f(t)dt = \int_{-\infty}^{-2} 0 dt + \int_{-2}^x \frac{1}{4} dt = \frac{1}{4}[t]_{-2}^x = \frac{1}{4}(x+2), \text{ for } -2 < x \leq 2$$

$$F(x) = \int_{-\infty}^x f(t)dt = \int_{-\infty}^{-2} 0 dt + \int_{-2}^2 \frac{1}{4} dt + \int_2^x 0 dt = \frac{1}{4}[t]_{-2}^2 = 1, \text{ for } x > 2$$

So, the distribution function $F(x)$ is given by:

$$F(x) = \begin{cases} 0 & , \text{ for } -\infty < x \leq -2 \\ \frac{1}{4}(x+2) & , \text{ for } -2 < x \leq 2 \\ 1 & , \text{ for } x > 2 \end{cases}$$

$$(i) \quad P(X < 1) = \int_{-\infty}^1 f(x)dx = \int_{-2}^1 \frac{1}{4} dx = \frac{1}{4}[x]_{-2}^1 = \frac{1}{4}(1+2) = \frac{3}{4}.$$

$$(ii) \quad P(|X| > 1) = 1 - P(|X| \leq 1) = 1 - P(-1 \leq X \leq 1)$$

$$= 1 - \int_{-1}^1 f(x)dx = 1 - \int_{-1}^1 \frac{1}{4} dx = 1 - \frac{1}{4}[x]_{-1}^1 = 1 - \frac{1}{2} = \frac{1}{2}.$$

$$(iii) \quad P(|X-1| \geq 2) = 1 - P(|X-1| < 2)$$

...(1)

$$\text{When } X-1 \geq 0: |X-1| < 2 \Rightarrow X-1 < 2 \Rightarrow X < 3.$$

$$\text{When } X-1 < 0: |X-1| < 2 \Rightarrow -(X-1) < 2 \Rightarrow X-1 > -2 \Rightarrow X > -1.$$

So, from (1), we get

$$P(|X-1| \geq 2) = 1 - P(-1 < X < 3) = 1 - \left\{ \int_{-1}^2 \frac{1}{4} dx + \int_2^3 0 dx \right\}$$

$$= 1 - \frac{1}{4}[x]_{-1}^2 = 1 - \frac{1}{4}(2+1) = \frac{1}{4}.$$

$$(iv) \quad P(2X+3 > 5) = P\left(X > \frac{5-3}{2}\right) = P(X > 1) = \int_1^{\infty} f(x)dx$$

$$= \int_1^2 f(x)dx = \int_1^2 \frac{1}{4} dx = \frac{1}{4}[x]_1^2 = \frac{1}{4}.$$

Example 13: Show that the function $f(x)$ given by

$$f(x) = \begin{cases} x & , 0 \leq x < 1 \\ k-x & , 1 \leq x \leq 2 \\ 0 & , \text{ elsewhere} \end{cases}$$

is a probability density function for a suitable value of the constant k . Construct the distribution function of the random variable X and compute the probability that the random variable X lies between $\frac{1}{2}$ and $\frac{3}{2}$.

(W.B.U.T. 2003, 2005, 2006, 2009, 2010)

Solution: The given function $f(x)$ is a possible probability density function (p.d.f.) if $f(x) \geq 0$

and $\int_{-\infty}^{\infty} f(x) dx = 1$.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Now,

$$\Rightarrow \int_{-\infty}^0 0 dx + \int_0^1 x dx + \int_1^2 (k-x) dx + \int_2^{\infty} 0 dx = 1$$

$$\Rightarrow \left[\frac{x^2}{2} \right]_0^1 + \left[kx - \frac{x^2}{2} \right]_1^2 = 1$$

$$\Rightarrow \frac{1}{2} + 2k - 2 - k + \frac{1}{2} = 1$$

$$k = 2.$$

$$\therefore f(x) = \begin{cases} x, & 0 \leq x < 1 \\ 2-x, & 1 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

Obviously, $f(x) \geq 0, \forall x$. So, $f(x)$ is a possible p.d.f. for $k = 2$.

The distribution function $F(x)$ is defined by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

For $-\infty < x < 0$: $F(x) = \int_{-\infty}^x f(t) dt = 0.$

For $0 \leq x < 1$: $F(x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^0 0 dt + \int_0^x t dt = \left[\frac{t^2}{2} \right]_0^x = \frac{x^2}{2}$

For $1 \leq x < 2$: $F(x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^0 0 dt + \int_0^1 t dt + \int_1^x (2-t) dt$

$$= \left[\frac{t^2}{2} \right]_0^1 + \left[2t - \frac{t^2}{2} \right]_1^x = \frac{1}{2} + \left(2x - \frac{x^2}{2} \right) - \left(2 - \frac{1}{2} \right)$$

$$= 2x - \frac{x^2}{2} - 1.$$

For $x \geq 2$: $F(x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^0 0 dt + \int_0^1 t dt + \int_1^2 (2-t) dt + \int_2^x 0 dt$

$$= \left[\frac{t^2}{2} \right]_0^1 + \left[2t - \frac{t^2}{2} \right]_1^2 = \frac{1}{2} + (4-2) - \left(2 - \frac{1}{2} \right) = 1.$$

So, the distribution function $F(x)$ of the random variable X is given by:

$$F(x) = \begin{cases} 0 & , -\infty < x < 0 \\ \frac{x^2}{2} & , 0 \leq x < 1 \\ 2x - \frac{x^2}{2} - 1 & , 1 \leq x < 2 \\ 1 & , x \geq 2 \end{cases}$$

$$\begin{aligned} \text{Now, } P\left(\frac{1}{2} < X < \frac{3}{2}\right) &= \int_{1/2}^{3/2} f(x) dx = \int_{1/2}^1 x dx + \int_1^{3/2} (2-x) dx \\ &= \left[\frac{x^2}{2}\right]_{1/2}^1 + \left[2x - \frac{x^2}{2}\right]_1^{3/2} \\ &= \left(\frac{1}{2} - \frac{1}{8}\right) + \left(3 - \frac{9}{8}\right) - \left(2 - \frac{1}{2}\right) = \frac{3}{4}. \end{aligned}$$

Example 14: Show that the function

$$f(x) = \begin{cases} |x| & , -1 < x < 1 \\ 0 & , \text{elsewhere} \end{cases}$$

is a possible probability density function and hence find the corresponding distribution function.

(W.B.U.T. 2006)

Solution: Observe that $f(x) \geq 0, \forall x$.

$$\begin{aligned} \text{Also, } \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^{-1} 0 dx + \int_{-1}^1 |x| dx + \int_1^{\infty} 0 dx = \int_{-1}^1 |x| dx \\ &= 2 \int_0^1 x dx \quad (\because |x| \text{ is an even function}) \\ &= 2 \left[\frac{x^2}{2}\right]_0^1 = 2 \cdot \frac{1}{2} = 1. \end{aligned}$$

Hence $f(x)$ is a possible probability density function.

The distribution function $F(x)$ is defined by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt.$$

$$\text{For } -\infty < x \leq -1: F(x) = \int_{-\infty}^x f(t) dt = 0.$$

$$\begin{aligned} \text{For } -1 < x < 0: F(x) &= \int_{-\infty}^x f(t) dt = \int_{-\infty}^{-1} 0 dt + \int_{-1}^x (-t) dt \quad [\because |t| = -t, \text{ for } t < 0] \\ &= -\left[\frac{t^2}{2}\right]_{-1}^x = -\frac{1}{2}(x^2 - 1) = \frac{1}{2}(1 - x^2). \end{aligned}$$

$$\begin{aligned} \text{For } 0 \leq x < 1: F(x) &= \int_{-\infty}^x f(t) dt = \int_{-\infty}^{-1} 0 dt + \int_{-1}^0 (-t) dt + \int_0^x t dt \\ &= -\left[\frac{t^2}{2}\right]_{-1}^0 + \left[\frac{t^2}{2}\right]_0^x = \frac{1}{2} + \frac{x^2}{2} = \frac{1}{2}(1 + x^2). \end{aligned}$$

For $x \geq 1$:

$$F(x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^{-1} 0 dt + \int_{-1}^1 |t| dt + \int_1^x 0 dx = 2 \int_0^1 t dt$$

$$= 2 \left[\frac{t^2}{2} \right]_0^1 = 2 \cdot \frac{1}{2} = 1.$$

So, the distribution function $F(x)$ of the random variable X is given by:

$$F(x) = \begin{cases} 0 & , -\infty < x \leq -1 \\ \frac{1}{2}(1-x^2) & , -1 < x < 0 \\ \frac{1}{2}(1+x^2) & , 0 \leq x < 1 \\ 1 & , x \geq 1 \end{cases}$$

Example 15: For what value of k , the function

$$f(x) = \begin{cases} ke^{-b(x-a)} & , a \leq x < \infty, a, b, x > 0 \\ 0 & , \text{elsewhere} \end{cases}$$

will be a probability density function?

(W.B.U.T. 2003)

Solution: The given function $f(x)$ is a possible probability density function if

$$f(x) \geq 0, \forall x \quad \text{and} \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

Now,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$\Rightarrow$

$$\int_{-\infty}^a 0 dx + \int_a^{\infty} ke^{-b(x-a)} dx = 1$$

$\Rightarrow$

$$k \lim_{B \rightarrow \infty} \int_a^B e^{-b(x-a)} dx = 1$$

$\Rightarrow$

$$k \lim_{B \rightarrow \infty} \left[\frac{e^{-b(x-a)}}{-b} \right]_a^B = 1$$

$\Rightarrow$

$$k \lim_{B \rightarrow \infty} \frac{1}{b} \{ 1 - e^{-b(B-a)} \} = 1$$

$\Rightarrow$

$$\frac{k}{b} = 1 \quad [\because e^{-b(B-a)} \rightarrow 0 \text{ as } B \rightarrow \infty, \text{ since } b > 0]$$

$\Rightarrow$

$$k = b.$$

$\therefore$

$$f(x) = \begin{cases} be^{-b(x-a)}, & a \leq x < \infty, a, b > 0 \\ 0 & , \text{elsewhere} \end{cases}$$

Obviously, $f(x) \geq 0, \forall x$. So, $f(x)$ will be a probability density function for $k = b$.

Example 16: The random variable X has the probability function

$$f(x) = k \quad , \quad \text{if } x = 0$$

$$= 3k \quad \text{if } x = 1$$

$$= 3k, \quad \text{if } x = 2$$

$$= 0, \quad \text{elsewhere}$$

Determine (i) value of k (ii) $P(X < 2)$ (iii) $P(X \leq 2)$ (iv) the smallest value of k for which $P(X \leq 1) \geq \frac{1}{2}$. (W.B.U.T. 2004)

Solution: Here the random variable X has the following probability distribution:

$X = x$	0	1	2
$f(x) = P(X = x)$	k	$2k$	$3k$

$$(i) \quad \Sigma f(x) = 1 \Rightarrow k + 2k + 3k = 1 \Rightarrow k = \frac{1}{6}$$

So, for $k = \frac{1}{6}$, $f(x) \geq 0$ and $\Sigma f(x) = 1$

$$(ii) \quad P(X < 2) = P(X = 0) + P(X = 1) = k + 2k = 3k = \frac{3}{6} = \frac{1}{2}$$

$$(iii) \quad P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) \\ = k + 2k + 3k = 6k = 1$$

$$(iv) \quad P(X \leq 1) \geq \frac{1}{2} \\ \Rightarrow P(X = 0) + P(X = 1) \geq \frac{1}{2} \\ \Rightarrow k + 2k \geq \frac{1}{2} \\ \Rightarrow k \geq \frac{1}{6}$$

Hence the smallest value of k is $\frac{1}{6}$.

Example 17: The length of the life of a tyre manufactured by a company follows a continuous distribution given by the density function

$$f(x) = \begin{cases} \frac{k}{x^3}, & 1000 \leq x \leq 1500 \\ 0, & \text{elsewhere} \end{cases}$$

Find k and find the probability that a randomly selected tyre would function for at least 1200 hours. (W.B.U.T. 2005)

Solution: The given function $f(x)$ is a possible probability density function if $f(x) \geq 0$, $\forall x$ and

$$\int_{-\infty}^{\infty} f(x) dx = 1.$$

Now,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_{-\infty}^{1000} 0 dx + \int_{1000}^{1500} \frac{k}{x^3} dx + \int_{1500}^{\infty} 0 dx = 1$$

$$\Rightarrow k \left[\frac{x^{-3+1}}{-3+1} \right]_{1000}^{1500} = 1$$

$$\Rightarrow \frac{k}{2} \cdot \frac{1}{10^4} \left(\frac{1}{100} - \frac{1}{225} \right) = 1$$

$$\Rightarrow k = 36 \times 10^5$$

$$\therefore f(x) = \begin{cases} 36 \times 10^5 x^{-3} & , \quad 1000 \leq x \leq 1500 \\ 0 & , \quad \text{elsewhere} \end{cases}$$

Obviously, $f(x) \geq 0, \forall x$.

$$\begin{aligned} \therefore P(X \geq 1200) &= \int_{1200}^{\infty} f(x) dx = \int_{1200}^{1500} 36 \times 10^5 x^{-3} dx + \int_{1500}^{\infty} 0 dx \\ &= 36 \times 10^5 \left[\frac{x^{-2}}{-2} \right]_{1200}^{1500} = 180 \left(\frac{1}{144} - \frac{1}{225} \right) = \frac{9}{20} \end{aligned}$$

This is the required probability.

Example 18: The life in hours of a certain type of electronic component of a computer follows a continuous distribution given by the density function:

$$f(x) = \begin{cases} \frac{k}{x^2} & , \quad x \geq 100 \\ 0 & , \quad x < 100 \end{cases}$$

Find k and determine the probability that all four such components in a computer will have to be replaced in the first 250 hours of its operation.

Solution: The given function $f(x)$ is a possible probability density function if $f(x) \geq 0, \forall x$ and $\int_{-\infty}^{\infty} f(x) dx = 1$.

$$\text{Now,} \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_{-\infty}^{100} 0 dx + \int_{100}^{\infty} kx^{-2} dx = 1$$

$$\Rightarrow k \lim_{B \rightarrow \infty} \int_{100}^B x^{-2} dx = 1$$

$$\Rightarrow k \lim_{B \rightarrow \infty} \left[-\frac{1}{x} \right]_{100}^B = 1$$

$$\Rightarrow k \lim_{B \rightarrow \infty} \left(\frac{1}{100} - \frac{1}{B} \right) = 1$$

$$\Rightarrow k = 100 \quad \left[\because \frac{1}{B} \rightarrow 0 \text{ as } B \rightarrow \infty \right]$$

$$\therefore f(x) = \begin{cases} 100x^{-2} & , \quad x \geq 100 \\ 0 & , \quad x < 100 \end{cases}$$

Obviously, $f(x) \geq 0, \forall x$. So, $f(x)$ is a possible probability density function for $k = 100$.

Let A_i denote the event 'the i th electronic component will last 250 hours or more', $i = 1, 2, 3, 4$.

$$\begin{aligned} \therefore P(A_i) &= \int_{250}^{\infty} f(x) dx = 100 \lim_{B \rightarrow \infty} \int_{250}^B x^{-2} dx \\ &= 100 \lim_{B \rightarrow \infty} \left[-\frac{1}{x} \right]_{250}^B = 100 \lim_{B \rightarrow \infty} \left(\frac{1}{250} - \frac{1}{B} \right) \\ &= \frac{100}{250} = \frac{2}{5}. \end{aligned}$$

$$\therefore P(\bar{A}_i) = 1 - P(A_i) = 1 - \frac{2}{5} = \frac{3}{5} \quad (i = 1, 2, 3, 4)$$

Since the events $\bar{A}_1, \bar{A}_2, \bar{A}_3$ and $\bar{A}_4$ are independent, the required probability is

$$\begin{aligned} P(\bar{A}_1 \bar{A}_2 \bar{A}_3 \bar{A}_4) &= P(\bar{A}_1)P(\bar{A}_2)P(\bar{A}_3)P(\bar{A}_4) \\ &= \left(\frac{3}{5}\right)^4 = \frac{81}{625}. \end{aligned}$$

Example 19: The p.d.f. of a random variable X is $f(x) = k(x-1)(2-x)$, for $1 \leq x \leq 2$. Determine

(i) the value of k (ii) the distribution function $F(x)$ (iii) $P\left(\frac{5}{4} \leq X \leq \frac{3}{2}\right)$. (W.B.U.T. 2007, 2010)

Solution: (i) The given function $f(x)$ is a possible probability density function (p.d.f.) if $f(x) \geq 0$,

$$\forall x \text{ and } \int_{-\infty}^{\infty} f(x) dx = 1.$$

$$\text{Now, } \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_{-\infty}^1 0 dx + \int_1^2 k(x-1)(2-x) dx + \int_2^{\infty} 0 dx = 1$$

$$\Rightarrow k \int_1^2 (3x - x^2 - 2) dx = 1$$

$$\Rightarrow k \left[\frac{3x^2}{2} - \frac{x^3}{3} - 2x \right]_1^2 = 1$$

$$\Rightarrow k \left[\left(6 - \frac{8}{3} - 4 \right) - \left(\frac{3}{2} - \frac{1}{3} - 2 \right) \right] = 1$$

$$\Rightarrow \frac{k}{6} = 1 \Rightarrow k = 6$$

$$\therefore f(x) = \begin{cases} 6(x-1)(2-x) & , \text{ for } 1 \leq x \leq 2 \\ 0 & , \text{ elsewhere} \end{cases}$$

Obviously, $f(x) \geq 0, \forall x$. So, $f(x)$ is a possible probability density function for $k = 6$.

(ii) The distribution function $F(x)$ of the random variable X is defined by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

For $-\infty < x < 1$: $F(x) = \int_{-\infty}^x f(t) dt = 0.$

For $1 \leq x < 2$:
$$F(x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^1 0 dt + \int_1^x 6(t-1)(2-t) dt = 6 \int_1^x (3t - t^2 - 2) dt$$

$$= 6 \left[\frac{3t^2}{2} - \frac{t^3}{3} - 2t \right]_1^x = 6 \left[\frac{3x^2}{2} - \frac{x^3}{3} - 2x - \left(\frac{3}{2} - \frac{1}{3} - 2 \right) \right]$$

$$= 5 - 12x + 9x^2 - 2x^3.$$

For $x \geq 2$: $F(x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^1 0 dt + \int_1^2 6(t-1)(2-t) dt + \int_2^{\infty} 0 dt = 1.$

So, the distribution function $F(x)$ of the random variable X is given by

$$F(x) = \begin{cases} 0 & , \quad -\infty < x < 1 \\ 5 - 12x + 9x^2 - 2x^3 & , \quad 1 \leq x < 2 \\ 1 & , \quad x \geq 2 \end{cases}$$

(iii) Using p.d.f.:

$$P\left(\frac{5}{4} \leq X \leq \frac{3}{2}\right) = \int_{5/4}^{3/2} f(x) dx = 6 \int_{5/4}^{3/2} (x-1)(2-x) dx$$

$$= 6 \left[\frac{3x^2}{2} - \frac{x^3}{3} - 2x \right]_{5/4}^{3/2}$$

$$= 6 \left[\frac{27}{8} - \frac{27}{24} - 3 - \left(\frac{75}{32} - \frac{125}{192} - \frac{5}{2} \right) \right] = \frac{11}{32}.$$

Alternative using d.f.:

$$P\left(\frac{5}{4} \leq X \leq \frac{3}{2}\right) = F\left(\frac{3}{2}\right) - F\left(\frac{5}{4}\right)$$

$$= 5 - 12 \cdot \frac{3}{2} + 9 \cdot \left(\frac{3}{2}\right)^2 - 2 \left(\frac{3}{2}\right)^3 - \left\{ 5 - 12 \cdot \frac{5}{4} + 9 \left(\frac{5}{4}\right)^2 - 2 \left(\frac{5}{4}\right)^3 \right\}$$

$$= \frac{11}{32}.$$

Example 20: For what values of a the function $f(x) = ax^{-2}u(x-a)$ is a possible probability density function of a r.v. X , where $u(x-a)$ is a unit step function. Construct the corresponding distribution function and compute $P(X \geq 10)$.

Solution: Since $u(x-a)$ is a unit step function, we have

$$u(x-a) = \begin{cases} 0 & , \quad \text{for } x < a \\ 1 & , \quad \text{for } x \geq a \end{cases} \quad \dots(1)$$

The given function $f(x)$ is a possible probability density function if $f(x) \geq 0$ and $\int_{-\infty}^{\infty} f(x)dx = 1$.

Now,
$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$\Rightarrow \int_{-\infty}^{\infty} ax^{-2}u(x-a)dx = 1$$

$$\Rightarrow a \int_a^{\infty} x^{-2}dx = 1 \quad [\text{By (1)}]$$

$$\Rightarrow a \lim_{B \rightarrow \infty} \int_a^B x^{-2}dx = 1$$

$$\Rightarrow a \lim_{B \rightarrow \infty} \left[\frac{x^{-1}}{-1} \right]_a^B = 1$$

$$\Rightarrow a \lim_{B \rightarrow \infty} \left(\frac{1}{a} - \frac{1}{B} \right) = 1$$

$$\Rightarrow a \cdot \frac{1}{a} = 1 \quad \left[\because \frac{1}{B} \rightarrow 0 \text{ as } B \rightarrow \infty \right]$$

$$\Rightarrow a > 0 \quad \left[\because f(x) \geq 0, \forall x \right]$$

So, $f(x)$ is a possible probability density function for all $a > 0$. The distribution function $F(x)$ is given by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt = \int_{-\infty}^x at^{-2}u(t-a)dt$$

$$= a \int_a^x t^{-2}dt \quad [\text{By (1)}]$$

$$= a \left[-\frac{1}{t} \right]_a^x = a \left(\frac{1}{a} - \frac{1}{x} \right) = 1 - \frac{a}{x}, \quad \text{where } x \geq a > 0.$$

$$F(x) = 0, \text{ elsewhere}$$

$$\therefore P(X \geq 10) = 1 - P(X < 10) = 1 - \left(1 - \frac{a}{10} \right) = \frac{a}{10}.$$

Example 21: Let $f(x) = ke^{-\alpha x}(1 - e^{-\alpha x})$, $x \geq 0$ where $\alpha > 0$. Find k such that $f(x)$ is a possible probability density function and determine the corresponding distribution function.

Solution: The given function $f(x)$ is a possible probability density function if $f(x) \geq 0$, $\forall x$ and

$$\int_{-\infty}^{\infty} f(x)dx = 1.$$

Now,
$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$\Rightarrow \int_{-\infty}^0 0dx + \int_0^{\infty} ke^{-\alpha x}(1 - e^{-\alpha x})dx = 1$$

$$\Rightarrow k \lim_{B \rightarrow \infty} \int_0^B (e^{-\alpha x} - e^{-2\alpha x})dx = 1$$

$$\Rightarrow k \lim_{B \rightarrow \infty} \frac{1}{2\alpha} [-2e^{-\alpha x} + e^{-2\alpha x}]_0^B = 1$$

$$\Rightarrow k \lim_{B \rightarrow \infty} \frac{1}{2\alpha} [-2e^{-\alpha B} + e^{-2\alpha B} + 1] = 1$$

$$\Rightarrow \frac{k}{2\alpha} = 1 \quad \left[\because e^{-\alpha B}, e^{-2\alpha B} \rightarrow 0 \text{ as } B \rightarrow \infty \right]$$

$$\Rightarrow k = 2\alpha$$

$$\therefore f(x) = 2\alpha e^{-\alpha x}(1 - e^{-\alpha x}), x \geq 0, \text{ where } \alpha > 0.$$

Obviously, $f(x) \geq 0, \forall x$. So, $f(x)$ is a possible probability density function for $k = 2\alpha$.

The distribution function $F(x)$ of the random variable X is defined by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt.$$

$$\text{For } -\infty < x < 0: \quad F(x) = \int_{-\infty}^x f(t) dt = 0.$$

For $x \geq 0$:

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(t) dt = \int_{-\infty}^0 0 dt + \int_0^x 2\alpha(e^{-\alpha t} - e^{-2\alpha t}) dt \\ &= 2\alpha \left[\frac{e^{-\alpha t}}{-\alpha} - \frac{e^{-2\alpha t}}{-2\alpha} \right]_0^x = \left[-2e^{-\alpha t} + e^{-2\alpha t} \right]_0^x \\ &= 1 - 2e^{-\alpha x} + e^{-2\alpha x} \end{aligned}$$

So, the distribution function $F(x)$ of the random variable X is given by:

$$F(x) = \begin{cases} 0 & , \quad -\infty < x < 0 \\ 1 - 2e^{-\alpha x} + e^{-2\alpha x} & , \quad x \geq 0, \alpha > 0 \end{cases}$$

Example 22: Suppose that the amount of money (in rupees) that a person has saved (negative savings indicate debt) is found to be a random phenomena described by the distribution function $F(x)$ as

$$F(x) = \begin{cases} \frac{1}{2} e^{-(x/5000)^2}, & x \leq 0 \\ 1 - \frac{1}{2} e^{-(x/5000)^2}, & x > 0 \end{cases}$$

- (i) Find the probability density function (p.d.f.) if it exists.
- (ii) What is the probability that the amount of savings possessed by a person will be (a) more than ₹ 5000 (b) less than - ₹ 5000 (c) between - ₹ 5000 and ₹ 5000 (d) equal to ₹ 5000?
- (iii) What is the conditional probability that the amount of savings possessed by a person will be (a) less than ₹ 10000 given that it is more than ₹ 5000 (b) more than ₹ 5000 given that it is less than ₹ 10000?

Solution: (i) The probability density function is given by:

$$f(x) = \frac{dF}{dx} = \begin{cases} \frac{-x}{25(10)^6} e^{-(x/5000)^2}, & x \leq 0 \\ \frac{x}{25(10)^6} e^{-(x/5000)^2}, & x > 0 \end{cases}$$

Here the random variable X denotes the savings of the person.

(ii) (a) $P(\text{amount of savings} > ₹ 5000)$

$$= P(X > 5000) = 1 - P(X \leq 5000)$$

$$= 1 - F(5000)$$

$$= 1 - \left\{ 1 - \frac{1}{2} e^{-(5000/5000)^2} \right\} = \frac{1}{2e}.$$

(b) $P(\text{Savings} < - ₹ 5000) = P(X < -5000)$

$$= F(-5000) = \frac{1}{2} e^{-(-5000/5000)^2} = \frac{1}{2e}.$$

(c) $P(\text{Savings lies between } -5000 \text{ and } 5000)$

$$= P(-5000 < X < 5000) = F(5000) - F(-5000)$$

$$= \left(1 - \frac{1}{2e} \right) - \frac{1}{2e}$$

[By (a) and (b)]

$$= 1 - \frac{1}{e}.$$

(d) $P(\text{Savings} = ₹ 5000) = P(X = 5000) = 0$, since for a continuous random variable the probability that it can take a particular value is zero.

(iii) (a) $P(\text{Savings is less than ₹ 10000 given that it is more than ₹ 5000})$

$$= P(X < 10000/X > 5000)$$

$$= \frac{P\{(X < 10000) \cap (X > 5000)\}}{P(X > 5000)}$$

[By definition]

$$= \frac{P(5000 < X < 10000)}{P(X > 5000)}$$

$$\text{Now, } P(5000 < X < 10000)$$

$$= F(10000) - F(5000)$$

$$= \left(1 - \frac{1}{2e^4} \right) - \left(1 - \frac{1}{2e} \right) = \frac{1}{2e} \left(1 - \frac{1}{e^3} \right)$$

$$\text{and } P(X > 5000) = \frac{1}{2e}.$$

$$\therefore \text{ Required probability} = \frac{\frac{1}{2e} \left(1 - \frac{1}{e^3} \right)}{\left(\frac{1}{2e} \right)} = 1 - \frac{1}{e^3}.$$

(b) $P(\text{Savings is more than ₹ 5000 given that it is less than ₹ 10000})$

$$= P(X > 5000/X < 10000) = \frac{P\{(X > 5000) \cap (X < 10000)\}}{P(X < 10000)}$$

$$= \frac{P(5000 < X < 10000)}{P(X < 10000)}$$

$$\text{Now, } P(5000 < X < 10000) = \frac{1}{2e} \left(1 - \frac{1}{e^3} \right)$$

and

$$P(X < 10000) = F(10000) = 1 - \frac{1}{2e^4}$$

$$\therefore \text{ Required probability} = \frac{\frac{1}{2e} \left(1 - \frac{1}{e^3} \right)}{\left(1 - \frac{1}{2e^4} \right)} = \frac{e^3 - 1}{2e^4 - 1}$$

Example 23: Determine the expected value of the number on a die when thrown.

Solution: Let X denotes the random variable which represents the number on a die when thrown. So, the probability distribution of X is

x_i	:	1	2	3	4	5	6
$P(x_i) = P(X = x_i)$	:	1/6	1/6	1/6	1/6	1/6	1/6

$$\begin{aligned} \therefore E(X) &= \sum x_i p(x_i) = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) \\ &= \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) \\ &= \frac{1}{6} \cdot \frac{1}{2} \cdot 6(6 + 1) = \frac{7}{2} \end{aligned}$$

This is the required expected value.

Example 24: An unbiased coin is tossed three times. Let X denotes the random variable which represents the number of tails appearing. Determine the probability distribution of X and hence find $E(X)$, $\text{Var}(X)$.

Solution: When a coin is tossed three times, then the sample space is

$$S = \{HHH, THH, HTH, HHT, HTT, THT, TTH, TTT\}.$$

Here X represents the random variable denoting the number of tails appearing. So, the probability distribution of X is

x_i	:	0	1	2	3
$P(x_i) = P(X = x_i)$	:	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$$\therefore E(X) = \sum x_i p(x_i) = 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8} = \frac{12}{8} = \frac{3}{2}$$

$$\begin{aligned} \text{Var}(X) &= \sum x_i^2 p(x_i) - \{E(X)\}^2 \\ &= 0^2 \times \frac{1}{8} + 1^2 \times \frac{3}{8} + 2^2 \times \frac{3}{8} + 3^2 \times \frac{1}{8} - \left(\frac{3}{2}\right)^2 = \frac{3}{4} \end{aligned}$$

Example 25: A die is tossed thrice. A success is 'getting 1 or 6' on a toss. Find the mean and variance of the number of successes.

Solution: Let S and F denote respectively the events 'one success' and 'one failure'.

$\therefore$

$$P(S) = P(\text{face 1 or 6}) = P(\text{face 1}) + P(\text{face 6})$$

$$= \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

$$P(F) = 1 - P(S) = 1 - \frac{1}{3} = \frac{2}{3}$$

Here the event space contains $2^3 = 8$ event points: $\{SSS, FSS, SFS, SSF, SFF, FSF, FFS, FFF\}$.

Let X denotes the random variable representing the number of successes for tossing a die thrice.

$$\therefore P(X=0) = P(FFF) = P(F) P(F) P(F) \quad (\text{Since the events are independent})$$

$$= \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{2}{3} = \frac{8}{27}$$

$$P(X=1) = P\{(SFF) \cup (FSF) \cup (FFS)\}$$

$$= P(SFF) + P(FSF) + P(FFS) \quad (\text{Since the events are m.e.})$$

$$= P(S) P(F) P(F) + P(F) P(S) P(F) + P(F) P(F) P(S)$$

($\because$ The events are independent)

$$= 3 \times \frac{1}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$$

$$P(X=2) = P\{(FSS) \cup (SFS) \cup (SSF)\}$$

$$= P(FSS) + P(SFS) + P(SSF) \quad (\text{Since the events are m.e.})$$

$$= P(F) P(S) P(S) + P(S) P(F) P(S) + P(S) P(S) P(F)$$

($\because$ The events are independent)

$$= 3 \times \frac{2}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{2}{9}$$

$$P(X=3) = P(SSS) = P(S) P(S) P(S) \quad (\because \text{The events are independent})$$

$$= \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{27}$$

The probability distribution of X is given below:

x_i	:	0	1	2	3
$p_i = P(X = x_i)$	:	8/27	4/9	2/9	1/27

$$\therefore \text{Mean} = E(X) = \sum x_i p_i = 0 \times \frac{8}{27} + 1 \times \frac{4}{9} + 2 \times \frac{2}{9} + 3 \times \frac{1}{27} = 1,$$

$$\text{Variance} = \text{Var}(X) = \sum x_i^2 p_i - \{E(X)\}^2 = 0^2 \times \frac{8}{27} + 1^2 \times \frac{4}{9} + 2^2 \times \frac{2}{9} + 3^2 \times \frac{1}{27} - 1^2$$

$$= \frac{4}{9} + \frac{8}{9} + \frac{3}{9} - 1 = \frac{2}{3}.$$

Example 26: Four defective bulbs are accidentally mixed with ten good bulbs. It is not possible to just look at a bulb and tell whether or not it is defective. Determine the probability distribution of the number of defective bulbs, if three bulbs are drawn at random without replacement from this lot. Also find the mean, variance and coefficient of variation of defective bulbs.

Solution: Let X denotes the random variable representing the number of defective bulbs in 3 drawn bulbs. Clearly X can take the values 0, 1, 2 or 3.

$$\text{Number of defective bulbs} = 4$$

Number of good bulbs = 10

Total number of bulbs = 14

$$P(X=0) = P(\text{no defective}) = P(\text{all 3 good ones})$$

$$= {}^{10}C_3 / {}^{14}C_3 = \frac{10!}{3!7!} \cdot \frac{3!11!}{14!} = \frac{10 \times 9 \times 8}{14 \times 13 \times 12}$$

$$= \frac{30}{91}$$

$$P(X=1) = P(1 \text{ defective and 2 good ones})$$

$$= {}^4C_1 {}^{10}C_2 / {}^{14}C_3$$

$$= 4 \cdot \frac{10!}{2!8!} \cdot \frac{3!11!}{14!} = 4 \cdot \frac{10 \times 9}{2} \cdot \frac{6}{14 \times 13 \times 12} = \frac{45}{91}$$

$$P(X=2) = P(2 \text{ defectives and 1 good one})$$

$$= {}^4C_2 {}^{10}C_1 / {}^{14}C_3$$

$$= \frac{4!}{2!2!} \cdot 10 \cdot \frac{3!11!}{14!} = \frac{6 \times 10 \times 6}{14 \times 13 \times 12} = \frac{15}{91}$$

$$P(X=3) = P(\text{all 3 defectives}) = {}^4C_3 / {}^{14}C_3 = 4 \cdot \frac{3!11!}{14!}$$

$$= \frac{4 \times 6}{14 \times 13 \times 12} = \frac{1}{91}$$

Thus the probability distribution of the random variable X is

x_i	:	0	1	2	3
$p_i = P(X=x_i)$	:	$\frac{30}{91}$	$\frac{45}{91}$	$\frac{15}{91}$	$\frac{1}{91}$

$\therefore$

$$\text{Mean} = E(X) = \sum x_i p_i$$

$$= 0 \times \frac{30}{91} + 1 \times \frac{45}{91} + 2 \times \frac{15}{91} + 3 \times \frac{1}{91} = \frac{78}{91}$$

$$\text{Variance} = \text{Var}(X) = \sum x_i^2 p_i - \{E(X)\}^2$$

$$= 0^2 \times \frac{30}{91} + 1^2 \times \frac{45}{91} + 2^2 \times \frac{15}{91} + 3^2 \times \frac{1}{91} - \left(\frac{78}{91}\right)^2$$

$$= \frac{114}{91} - \left(\frac{78}{91}\right)^2 = \frac{4290}{8281}$$

$\therefore$

$$\text{S.D.} = +\sqrt{\text{Var}(X)} = 0.72$$

$$\text{Coefficient of variation} = \frac{\text{S.D.}}{\text{Mean}} \times 100 = 0.72 \times \frac{91}{78} \times 100 = 84\%$$

Example 27: The mean and s.d. of marks of 70 students were found to be 65 and 5.2 respectively. Later it was detected that the value 85 was recorded wrongly and therefore it was removed from the data set. Then find the mean and s.d. for the remaining 69 students. (W.B.U.T. 2008)

Solution: Calculation of mean

$$\bar{X} = \frac{\Sigma X}{N} \Rightarrow N\bar{X} = \Sigma X$$

Given,

$$N = 70, \bar{X} = 65$$

$\therefore$

$$\Sigma X = 70 \times 65 = 4550$$

$\therefore$ Mean of the remaining 69 students

$$= \frac{\Sigma X - 85}{69} = \frac{4550 - 85}{69} = \frac{4465}{69} = 64.71$$

Calculation of S.D.

$$\sigma^2 = \frac{\Sigma X^2}{N} - (\bar{X})^2$$

Given

$$N = 70, \bar{X} = 65 \text{ and } \sigma = 5.2$$

$\therefore$

$$(5.2)^2 = \frac{\Sigma X^2}{70} - (65)^2$$

$\Rightarrow$

$$\Sigma X^2 = 70\{(5.2)^2 + (65)^2\}$$

$\Rightarrow$

$$\frac{\Sigma X^2 - (85)^2}{69} = 4208.95$$

$\therefore$ S.D. of the remaining 69 students

$$= \sqrt{4208.95 - (64.71)^2} = 4.64.$$

Example 28: The arithmetic mean and standard deviation of a series of 20 items were calculated by a student as 25 and 6 respectively. But while calculating an item 14 was misread as 40. Find the correct arithmetic mean and standard deviation.

Solution: Calculation of correct mean

$$\bar{X} = \frac{\Sigma X}{N} \Rightarrow \Sigma X = N\bar{X}$$

Given,

$$N = 20, \bar{X} = 25$$

$\therefore$

$$\Sigma X = 20 \times 25 = 500$$

Correct

$$\Sigma X = 500 - 40 + 14 = 474.$$

Correct

$$\bar{X} = \frac{474}{20} = 23.7$$

Calculation of correct standard deviation

$$\sigma^2 = \frac{\Sigma X^2}{N} - (\bar{X})^2$$

Given,

$$N = 20, \bar{X} = 25 \text{ and } \sigma = 6.$$

$$\begin{aligned}
 \therefore 6^2 &= \frac{\Sigma X^2}{20} - (25)^2 \\
 \Rightarrow \Sigma X^2 &= 20(36 + 625) = 13220 \\
 \text{Correct } \Sigma X^2 &= 13220 - (40)^2 + (14)^2 = 11816 \\
 \therefore \text{Correct } \sigma &= \sqrt{\frac{\text{Correct } \Sigma X^2}{N} - (\text{Correct } \bar{X})^2} \\
 &= \sqrt{\frac{11816}{20} - (23.7)^2} = 5.4.
 \end{aligned}$$

Hence the correct mean is 23.7 and correct standard deviation is 5.4.

Example 29: Two persons throw an ordinary die alternately and the first who throws 6 is to receive ₹ 121. Find their expectations.

Solution: Probability of throwing 6 with a die = $\frac{1}{6}$

Probability of not throwing 6 with a die = $1 - \frac{1}{6} = \frac{5}{6}$.

First man wins if he throws 6 in any of the 1st, 3rd, 5th, ... throws (and the second man does not throw 6 in 2nd, 4th, ... throws).

Probability of throwing 6 in the first throw = $\frac{1}{6}$

Probability of throwing 6 in the third throw = $\frac{5}{6} \cdot \frac{5}{6} \cdot \frac{1}{6}$

Probability of throwing 6 in the fifth throw = $\frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{1}{6} \dots$

Hence probability of first man's success

$$\begin{aligned}
 &= \frac{1}{6} + \left(\frac{5}{6}\right)^2 \frac{1}{6} + \left(\frac{5}{6}\right)^4 \frac{1}{6} + \dots \text{to } \infty \\
 &= \frac{1}{6} \left\{ 1 + \left(\frac{5}{6}\right)^2 + \left(\frac{5}{6}\right)^4 + \dots \text{to } \infty \right\} \\
 &= \frac{1}{6} \left\{ 1 - \left(\frac{5}{6}\right)^2 \right\}^{-1} = \frac{1}{6} \cdot \frac{36}{11} = \frac{6}{11}.
 \end{aligned}$$

So, probability of second man's success = $1 - \frac{6}{11} = \frac{5}{11}$.

Hence first man's expectation = $xP(X=x) = 121P(X=121) = 121 \times \frac{6}{11} = ₹ 66$

Second man's expectation = $121 P(Y=121) = 121 \times \frac{5}{11} = ₹ 55$.

Example 30: The probability density function of a continuous distribution is given by:

$$f(x) = \frac{3}{4}x(2-x), 0 < x < 2.$$

Compute mean and variance.

(W.B.U.T. 2004, 2007, 2008, 2009)

Solution: Let X be the continuous random variable having p.d.f.:

$$f(x) = \frac{3}{4}x(2-x), 0 < x < 2$$

$$\therefore \text{Mean} = E(x)$$

$$\begin{aligned} &= \int_{-\infty}^{\infty} xf(x) dx = \int_0^2 x \cdot \frac{3}{4}x(2-x) dx \\ &= \frac{3}{4} \left[\frac{2}{3}x^3 - \frac{x^4}{4} \right]_0^2 = \frac{3}{4} \left(\frac{16}{3} - 4 \right) = 4 - 3 = 1. \end{aligned}$$

$$\begin{aligned} \text{Now, } E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^2 x^2 \cdot \frac{3}{4}x(2-x) dx = \frac{3}{4} \left[2 \cdot \frac{x^4}{4} - \frac{x^5}{5} \right]_0^2 \\ &= \frac{3}{4} \left(8 - \frac{32}{5} \right) = \frac{3}{4} \cdot \frac{8}{5} = \frac{6}{5}. \end{aligned}$$

$$\therefore \text{Var}(X) = E(X^2) - \{E(X)\}^2 = \frac{6}{5} - 1^2 = \frac{1}{5}.$$

Example 31: Let X be a random variable. Then show that

$$|E(X)| \leq E(|X|).$$

Solution: We know that $X \leq |X|$

$$\Rightarrow X - |X| \leq 0$$

$$\Rightarrow E(X - |X|) \leq 0$$

$$\Rightarrow E(X) - E(|X|) \leq 0$$

$$\Rightarrow E(X) \leq E(|X|) \quad \dots(1)$$

$$\text{Again, } -X \leq |X|$$

$$\Rightarrow E(-X) \leq E(|X|)$$

$$\Rightarrow -E(X) \leq E(|X|) \quad \dots(2)$$

From (1) and (2), we get

$$|E(X)| \leq E(|X|).$$

Example 32: A continuous variate X has the p.d.f.

$$\begin{aligned} f(x) &= x/2, 0 \leq x \leq 1 \\ &= 1/2, 1 < x \leq 2 \\ &= (3-x)/2, 2 < x \leq 3 \end{aligned}$$

Find (i) mean of X

(ii) Variance of X .

(W.B.U.T. 2002)

Solution:

(i)

$$\begin{aligned}
 \text{Mean} = E(X) &= \int_{-\infty}^{\infty} x f(x) dx \\
 &= \int_{-\infty}^0 x \cdot 0 dx + \int_0^1 x \cdot \frac{x}{2} dx + \int_1^2 x \cdot \frac{1}{2} dx + \int_2^3 x \cdot \frac{(3-x)}{2} dx + \int_3^{\infty} x \cdot 0 dx \\
 &= \frac{1}{2} \left[\frac{x^3}{3} \right]_0^1 + \frac{1}{2} \left[\frac{x^2}{2} \right]_1^2 + \frac{1}{2} \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_2^3 \\
 &= \frac{1}{6} + \frac{1}{2} \left(2 - \frac{1}{2} \right) + \frac{1}{2} \left(\frac{27}{2} - 9 - 6 + \frac{8}{3} \right) \\
 &= \frac{1}{6} + \frac{3}{4} - \frac{15}{2} + \frac{97}{12} = \frac{1}{12} (2 + 9 - 90 + 97) = \frac{3}{2}.
 \end{aligned}$$

(ii)

$$\begin{aligned}
 E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx \\
 &= \int_0^1 x^2 \cdot \frac{x}{2} dx + \int_1^2 x^2 \cdot \frac{1}{2} dx + \int_2^3 x^2 \cdot \frac{(3-x)}{2} dx \\
 &= \frac{1}{2} \left[\frac{x^4}{4} \right]_0^1 + \frac{1}{2} \left[\frac{x^3}{3} \right]_1^2 + \frac{1}{2} \left[x^3 - \frac{x^4}{4} \right]_2^3 \\
 &= \frac{1}{8} + \frac{1}{2} \cdot \frac{7}{3} + \frac{1}{2} \left[27 - \frac{81}{4} - 8 + 4 \right] = \frac{1}{8} + \frac{7}{6} + \frac{23}{2} - \frac{81}{8} \\
 &= -10 + \frac{38}{3} = \frac{8}{3}.
 \end{aligned}$$

$$\therefore \text{Var}(X) = E(X^2) - \{E(X)\}^2 = \frac{8}{3} - \left(\frac{3}{2} \right)^2 = \frac{5}{12}.$$

Example 33: For the p.d.f.:

$$f(x) = \begin{cases} \frac{2(b+x)}{b(a+b)} & , -b \leq x \leq 0 \\ \frac{2(a-x)}{a(a+b)} & , 0 \leq x \leq a \end{cases}$$

Find the mean and variance.

(W.B.U.T. 2003)

Solution: Let X be the continuous random variable having p.d.f.:

$$f(x) = \begin{cases} \frac{2(b+x)}{b(a+b)} & , -b \leq x \leq 0 \\ \frac{2(a-x)}{a(a+b)} & , 0 \leq x \leq a \end{cases}$$

$$\begin{aligned}
 \therefore \text{Mean} = E(X) &= \int_{-\infty}^{\infty} x f(x) dx = \int_{-b}^0 x \left\{ \frac{2(b+x)}{b(a+b)} \right\} dx + \int_0^a x \left\{ \frac{2(a-x)}{a(a+b)} \right\} dx \\
 &= \frac{2}{b(a+b)} \left[b \frac{x^2}{2} + \frac{x^3}{3} \right]_{-b}^0 + \frac{2}{a(a+b)} \left[a \frac{x^2}{2} - \frac{x^3}{3} \right]_0^a \\
 &= \frac{2}{b(a+b)} \left[-\frac{b^3}{2} + \frac{b^3}{3} \right] + \frac{2}{a(a+b)} \left[\frac{a^3}{2} - \frac{a^3}{3} \right] \\
 &= \frac{1}{3(a+b)} (-b^2 + a^2) = \frac{1}{3}(a-b)
 \end{aligned}$$

$$\begin{aligned}
 \text{Now, } E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx = \int_{-b}^0 x^2 \left\{ \frac{2(b+x)}{b(a+b)} \right\} dx + \int_0^a x^2 \left\{ \frac{2(a-x)}{a(a+b)} \right\} dx \\
 &= \frac{2}{b(a+b)} \left[b \frac{x^3}{3} + \frac{x^4}{4} \right]_{-b}^0 + \frac{2}{a(a+b)} \left[a \frac{x^3}{3} - \frac{x^4}{4} \right]_0^a \\
 &= \frac{2}{b(a+b)} \left[\frac{b^4}{3} - \frac{b^4}{4} \right] + \frac{2}{a(a+b)} \left[\frac{a^4}{3} - \frac{a^4}{4} \right] = \frac{1}{6(a+b)} (b^3 + a^3) \\
 &= \frac{1}{6} (a^2 - ab + b^2)
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Var}(X) &= E(X^2) - \{E(X)\}^2 = \frac{1}{6} (a^2 - ab + b^2) - \frac{1}{9} (a-b)^2 \\
 &= \frac{1}{18} (a^2 + ab + b^2).
 \end{aligned}$$

Example 34: A bag contains a white and b black balls. If n balls are drawn at random, compute the expected (or mean) value of the number of white balls drawn.

Solution: Let X_i denotes a random variable associated with i th drawn. We define:

$$\begin{aligned}
 X_i &= 1 \text{ if the } i\text{th ball drawn is white} \\
 &= 0 \text{ if the } i\text{th ball drawn is black,}
 \end{aligned}$$

where $i = 1, 2, \dots, n$.

Then the number of white balls among n balls drawn is given by the random variable

$$X = \sum_{i=1}^n X_i \quad (\because X_i = 1 \text{ only if } i\text{th ball drawn is white})$$

$$\text{Now, } P(X_i = 1) = P(\text{of drawing white ball}) = \frac{a}{a+b}$$

$$P(X_i = 0) = P(\text{of drawing black ball}) = \frac{b}{a+b}.$$

$$\therefore E(X_i) = 1 \cdot P(X_i = 1) + 0 \cdot P(X_i = 0) = \frac{a}{a+b}.$$

$$\therefore E(X) = E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E(X_i) = \frac{an}{a+b}.$$

Example 35: An electronic device has random life span and is associated with a random variable X (in units of 1000 hours) with the following p.d.f.:

$$f(x) = \begin{cases} e^{-x} & , x > 0 \\ 0 & , x \leq 0 \end{cases}$$

If the cost of one such item is ₹ 200 and the manufacturer sells each item for ₹ 600, but guarantees a total refund if $X \leq 0.9$, find the manufacturer's expected profit per item (assume $e^{-0.9} = 0.4066$).

Solution: $P(X \leq 0.9) = \int_0^{0.9} f(x) dx = \int_0^{0.9} e^{-x} dx = \left[-e^{-x}\right]_0^{0.9} = 1 - e^{-0.9}$

$\therefore P(X > 0.9) = 1 - P(X \leq 0.9) = 1 - (1 - e^{-0.9}) = e^{-0.9}$

$\therefore$ Required expected profit

$$\begin{aligned} &= -200 P(X \leq 0.9) + (600 - 200) P(X > 0.9) \\ &= -200(1 - e^{-0.9}) + 400e^{-0.9} = 600e^{-0.9} - 200 \\ &= 600 \times 0.4066 - 200 = ₹ 43.96. \end{aligned}$$

Example 36: A man buys a ticket in a lottery that sells 100 tickets and that gives 5 prizes of ₹ 200, 8 prizes of ₹ 100 and 20 prizes of ₹ 10. How much should the man be willing to pay for a ticket in this lottery? Assume that all tickets are equally likely to win the prize.

Solution: Let the man's gain be expressed by the random variable X . By question, X takes the value 200 five times, 100 eight times and 10 twenty times. It is assumed that all tickets are equally likely to win the prize.

$$\begin{aligned} \therefore E(X) &= 5 \times 200 \times \frac{1}{100} + 8 \times 100 \times \frac{1}{100} + 20 \times 10 \times \frac{1}{100} \\ &= 10 + 8 + 2 = 20. \end{aligned}$$

So the man should pay ₹ 20 for a single ticket.

Example 37: A bowl contains 10 chips, of which 8 are marked ₹ 200 each and 2 are marked ₹ 500 each. Let a person draw 3 chips at random from the bowl without replacement. If the person is to receive the sum of resulting amounts marked on the drawn chips, find his expectation.

Solution: There are two types of chips in the bowl denoted as follows:

A : Chips marked ₹ 200 each

B : Chips marked ₹ 500 each

3 chips out of 10 chips can be drawn (ignoring the order) in the following ways:

AAA, AAB, ABB

($\because$ There are only two chips of type B)

Let X be the random variable which represents the sum of the amount marked on the drawn three chips. Then X takes the values 600, 900, 1200.

$$\begin{aligned} \therefore P(X = 600) &= P(AAA) = \frac{{}^8C_3}{{}^{10}C_3} = \frac{8!}{3!5!} \cdot \frac{3!7!}{10!} \\ &= \frac{7 \times 6}{10 \times 9} = \frac{7}{15}, \end{aligned}$$

$$P(X = 900) = P(AAB) = {}^8C_2 \times {}^2C_1 / {}^{10}C_3 = \frac{2!8!}{2!6!} \cdot \frac{3!7!}{10!}$$

$$= \frac{6 \times 7}{10 \times 9} = \frac{7}{15},$$

$$P(X = 1200) = P(ABB) = {}^8C_1 \times {}^2C_2 / {}^{10}C_3 = 8 \times 1 \times \frac{3!7!}{10!}$$

$$= \frac{6}{10 \times 9} = \frac{1}{15}.$$

Hence,

$$E(X) = 600P(X = 600) + 900P(X = 900) + 1200P(X = 1200)$$

$$= 600 \times \frac{7}{15} + 900 \times \frac{7}{15} + 1200 \times \frac{1}{15}$$

$$= 280 + 420 + 80 = ₹ 780$$

∴ The expectation of the person is ₹ 780.

Example 38: The demand for a new product of a company is assumed to be a continuous random variable with the following p.d.f.:

$$f(x) = \begin{cases} \frac{x}{a^2} e^{-x^2/2a^2} & , \quad x \geq 0 \\ 0 & , \quad x < 0 \end{cases}$$

Find the mean and variance of this random variable and also find the probability that the demand will exceed a .

Solution: Now, $E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x \cdot \frac{x}{a^2} e^{-x^2/2a^2} dx$

$$= \lim_{B \rightarrow \infty} \int_0^B x \cdot \frac{x}{a^2} e^{-x^2/2a^2} dx \quad \dots(1)$$

By integration by parts,

$$\begin{aligned} \int x \cdot \frac{x}{a^2} e^{-x^2/2a^2} dx &= x \int \frac{x}{a^2} e^{-x^2/2a^2} dx - \int \left\{ \left(\frac{dx}{dx} \right) \int \frac{x}{a^2} e^{-x^2/2a^2} dx \right\} dx \\ &= -x e^{-x^2/2a^2} + \int e^{-x^2/2a^2} dx \end{aligned}$$

So, from (1):

$$E(X) = - \lim_{B \rightarrow \infty} [x e^{-x^2/2a^2}]_0^B + \lim_{B \rightarrow \infty} \int_0^B e^{-x^2/2a^2} dx$$

Now, $\lim_{B \rightarrow \infty} B e^{-B^2/2a^2} = \lim_{B \rightarrow \infty} \frac{B}{e^{B^2/2a^2}} \left(\frac{\infty}{\infty} \right)$

$$= \lim_{B \rightarrow \infty} \frac{1}{\frac{B}{a^2} e^{B^2/2a^2}} = 0.$$

$$\therefore E(X) = \lim_{B \rightarrow \infty} \int_0^B e^{-x^2/2a^2} dx = \lim_{B \rightarrow \infty} \frac{a}{\sqrt{2}} \int_0^{B^2/2a^2} e^{-u} u^{\frac{1}{2}-1} du \left[\text{Put } u = \frac{x^2}{2a^2} \right]$$

$$\therefore E(X) = \frac{a}{\sqrt{2}} \int_0^\infty e^{-u} u^{\frac{1}{2}-1} du = \frac{a}{\sqrt{2}} \Gamma\left(\frac{1}{2}\right) = a\sqrt{\frac{\pi}{2}}$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^\infty x^2 f(x) dx = \int_0^\infty x^2 \cdot \frac{x}{a^2} e^{-x^2/2a^2} dx \\ &= \lim_{B \rightarrow \infty} \int_0^B x^2 \cdot \frac{x}{a^2} e^{-x^2/2a^2} dx \quad \dots(2) \end{aligned}$$

By integration by parts,

$$\begin{aligned} \int x^2 \cdot \frac{x}{a^2} e^{-x^2/2a^2} dx &= x^2 \int \frac{x}{a^2} e^{-x^2/2a^2} dx - \int \left\{ \left(\frac{dx^2}{dx} \right) \int \frac{x}{a^2} e^{-x^2/2a^2} dx \right\} dx \\ &= -x^2 e^{-x^2/2a^2} + 2 \int x e^{-x^2/2a^2} dx \\ &= -x^2 e^{-x^2/2a^2} - 2a^2 e^{-x^2/2a^2} \end{aligned}$$

So, from (2):

$$\begin{aligned} E(X^2) &= - \lim_{B \rightarrow \infty} \left[x^2 e^{-x^2/2a^2} \right]_0^B - 2a^2 \lim_{B \rightarrow \infty} \left[e^{-x^2/2a^2} \right]_0^B \\ &= - \lim_{B \rightarrow \infty} \frac{B^2}{e^{B^2/2a^2}} - 2a^2 \lim_{B \rightarrow \infty} \left[e^{-B^2/2a^2} - 1 \right] \\ &= -0 + 2a^2 \text{ [by L' Hospital rule and since } \lim_{B \rightarrow \infty} e^{-B^2/2a^2} = 0] \\ &= 2a^2. \end{aligned}$$

$$\therefore \text{Var}(X) = E(X^2) - \{E(X)\}^2 = 2a^2 - \frac{\pi}{2} a^2 = (4 - \pi) \frac{a^2}{2}.$$

$$\therefore P(\text{Demand will exceed } a) = P(X > a) = 1 - P(X \leq a)$$

$$\begin{aligned} &= 1 - \int_{-\infty}^a f(x) dx = 1 - \int_0^a \frac{x}{a^2} e^{-x^2/2a^2} dx = 1 + \left[e^{-x^2/2a^2} \right]_0^a \\ &= e^{-1/2} = \frac{1}{\sqrt{e}}. \end{aligned}$$

MULTIPLE CHOICE QUESTIONS

$$1. \text{ If } P(X=x) = \begin{cases} \frac{x}{21}, & \text{for } x=1, 2, \dots, 6 \\ 0, & \text{otherwise} \end{cases}$$

then $P(X=2 \text{ or } 3)$ is

(a) 2/21

(b) 3/21

(c) 4/21

(d) 5/21.

2. If for a random variable X , $\text{Var}(X) = 1$, then $\text{Var}(2X + 3)$ is
 (a) 1 (b) 2 (c) 4 (d) none of these.
3. The A.M. of the first n natural numbers is
 (a) $n(n+1)/2$ (b) $(n+1)/2$ (c) $n/2$ (d) n .
4. A random variable X has the following p.d.f.:

$$f(x) = \begin{cases} k & , -2 < x < 2 \\ 0 & , \text{elsewhere} \end{cases}$$

Then the value of the constant k is

- (a) $\frac{1}{8}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{12}$.
 (W.B.U.T. 2007, 2012)
5. The variance of a random variable X is
 (a) $\{E(X)\}^2$ (b) $E(X^2)$ (c) $E(X^2) - \{E(X)\}^2$ (d) $E(X^2) - E(X)$.
 (W.B.U.T. 2006, 2012)
6. The probability $P(a \leq X \leq b)$ [where $F(x)$ is the distribution function of the random variable X] is defined by
 (a) $F(b) - F(a)$ (b) $F(b) + F(a)$ (c) $F(a) - F(b)$ (d) $F(a) F(b)$.
 (W.B.U.T. 2006, 2008, 2009)
7. If X and Y are independent random variables, then
 (a) $E(XY) = E(X) + E(Y)$ (b) $E(XY) = E(X) - E(Y)$ (c) $E(XY) = E(X) E(Y)$ (d) $E(XY) = E(X)/E(Y)$.
 (W.B.U.T. 2006)

8. $f(x) = \begin{cases} kx(1-x) & , 0 < x < 1 \\ 0 & , \text{elsewhere} \end{cases}$

shall be a probability density function if $k =$

- (a) 1 (b) 2 (c) 3 (d) 6.
9. A random variable X has the following p.d.f.:
- $$f(x) = \begin{cases} e^{-x} & , x \geq 0 \\ 0 & , \text{elsewhere} \end{cases}$$
- Then the value of $E(X)$ is
 (a) 0 (b) 1 (c) 2 (d) -1.
 (W.B.U.T. 2007)
10. If $F(x)$ is the distribution function of a random variable, then
 (a) $F(x)$ is continuous at all points (b) $F(x)$ is monotonic decreasing function
 (c) $F(-\infty) = 1$ (d) $F(+\infty) = 1$.
 (W.B.U.T. 2008)
11. If X is a discrete random variable, then
 (a) $E(|X|) \leq |E(X)|$ (b) $E(|X|) \geq |E(X)|$ (c) $E(|X|) = |E(X)|$ (d) none of these.
 (W.B.U.T. 2008)
12. Let X and Y be two random variables such that $Y = a + bX$ where a and b are constants. Then, $\text{Var}(Y)$ is
 (a) $b^2 \text{Var}(X)$ (b) $\text{Var}(X)$ (c) $a^2 \text{Var}(X)$ (d) $(b/a) \text{Var}(X)$.
 (W.B.U.T. 2008)

13. The random variable X has distribution function F . If $F(0) = \frac{1}{6}$, $F(1) = \frac{1}{2}$, $F(3) = \frac{3}{4}$, then $P(0 < X \leq 3)$ is
 (a) $3/4$ (b) $1/3$ (c) $7/12$ (d) none of these.
 [Hint: $P(0 < X \leq 3) = F(3) - F(0)$] (W.B.U.T. 2010)
14. The distribution function $F(x)$ of a random variable X is given by [where $-\infty < X < \infty$]
 (a) $P(-\infty < X < \infty)$ (b) $P(-\infty < X \leq x)$ (c) $P(-\infty \leq X < \infty)$ (d) none of these.
 (W.B.U.T. 2010)
15. In a probability distribution of a random variable X , the sum of the probabilities is always
 (a) 0 (b) less than 1 (c) 1 (d) none of these.
16. Which of the following can be taken as probability distribution of a discrete random variable for the range 1, 2, 3, 4:
 (a) $p(x) = \frac{x^2}{15}$ (b) $p(x) = \frac{x-2}{2}$ (c) $p(x) = \frac{x+2}{18}$ (d) none of these.

ANSWERS

- | | | | | | |
|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (c) | 3. (b) | 4. (c) | 5. (c) | 6. (a) |
| 7. (c) | 8. (d) | 9. (b) | 10. (d) | 11. (b) | 12. (a) |
| 13. (c) | 14. (b) | 15. (c) | 16. (c) | | |

PROBLEMS

- Determine the probability distribution of the number of doublets in four throws of a pair of fair dice.
- Two bad eggs are accidentally mixed with ten good eggs. It is not possible to just look at an egg and tell whether or not it is bad. Determine the probability distribution of the number of bad eggs, if three eggs are drawn at random without replacement from this lot.
- Determine the distribution function of the total number of heads turn up in three tosses of a fair coin.
- A random variable X has the following probability distribution:

$X = x$	:	0	1	2	3	4	5	6	7	8
$P(X = x)$	:	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$	$15k$	$17k$

 Find (i) the value of k , (ii) $P(X < 3)$, $P(X \geq 3)$, $P(2 \leq X < 5)$, (iii) the smallest value of x for which $P(X \leq x) > 0.5$.

5. A random variable X gives measurements between 0 and 1 with a probability density function:

$$f(x) = \begin{cases} 12x^3 - 21x^2 + 10x & , \quad 0 \leq x \leq 1 \\ 0 & , \quad \text{elsewhere} \end{cases}$$

Find (i) $P\left(X \leq \frac{1}{2}\right)$, $P\left(X > \frac{1}{2}\right)$ and (ii) a number k such that $P(X \leq k) = \frac{1}{2}$.

6. A function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} 0 & , \quad x < 2 \\ \frac{1}{18}(2x+3) & , \quad 2 \leq x \leq 4 \\ 0 & , \quad x > 4 \end{cases}$$

Prove that it is a possible probability density function. Determine the probability that a variate having this density will fall in the interval $2 \leq x \leq 3$.

7. Let X is a continuous random variable with the following probability density function:

$$f(x) = \begin{cases} ax & , 0 \leq x < 2 \\ 2a & , 2 \leq x < 4 \\ -ax + 6a & , 4 \leq x < 6 \\ 0 & , \text{elsewhere} \end{cases}$$

Find a and mean value of X .

8. A function $f(x)$ is defined as follows:

$$\begin{aligned} f(x) &= \frac{1}{16}(3+x)^2, & -3 \leq x < -1 \\ &= \frac{1}{16}(2-6x^2), & -1 \leq x < 1 \\ &= \frac{1}{16}(3-x)^2, & 1 \leq x \leq 3 \\ &= 0, & \text{elsewhere} \end{aligned}$$

Show that it is a possible probability density function and prove that the mean of a random variable having this density is zero.

9. Find the variance and standard deviation for the following discrete distribution:

$X = x$	:	8	12	16	20	24
$P(X = x)$	:	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{1}{12}$

10. The probability density function of a continuous random variable is given below:

$$f(x) = \begin{cases} y_0 x(2-x) & , 0 \leq x \leq 2 \\ 0 & , \text{elsewhere} \end{cases}$$

Find (i) the value of y_0 , (ii) mean and variance of the distribution.

11. The probability density function of a continuous random variable is given below:

$$p(x) = y_0 e^{-|x|}, -\infty < x < \infty$$

Find (i) the value of y_0 , (ii) mean and variance of the distribution.

12. A function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} \frac{1}{2}(x+1) & , -1 < x < 1 \\ 0 & , \text{elsewhere} \end{cases}$$

Show that it is a possible probability density function and compute the mean and variance of a random variable having this density.

13. Five balls are drawn from an urn containing 4 white and 6 black balls. Find the probability distribution of the number of white balls drawn without replacement. (W.B.U.T. 2004)

14. If the probability density function of a random variable X is given by

$$f(x) = ce^{-(x-2)^2/2}, -\infty < x < \infty,$$

find the value of c , the expectation and variance of the distribution.

(W.B.U.T. 2007)

ANSWERS

1.	$X = x_i$	:	0	1	2	3	4
	$P(X = x_i)$	:	$\frac{625}{1296}$	$\frac{500}{1296}$	$\frac{150}{1296}$	$\frac{20}{1296}$	$\frac{1}{1296}$

Hints: There can be 6 doublets, namely (1, 1), (2, 2), (3, 3), (4, 4), (5, 5) and (6, 6).

Probability of throwing a doublet in one throw = $\frac{6}{6^2} = \frac{1}{6}$.

Probability of not throwing a doublet in one throw = $1 - \frac{1}{6} = \frac{5}{6}$.

$$P(X=0) = \left(\frac{5}{6}\right)^4, P(X=1) = {}^4C_1 \left(\frac{1}{6}\right) \left(\frac{5}{6}\right)^3, P(X=2) = {}^4C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^2, P(X=3) = {}^4C_3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right),$$

$$P(X=4) = \left(\frac{1}{6}\right)^4.$$

2.	$X = x_i$	:	0	1	2
	$P(X = x_i)$	:	$\frac{12}{22}$	$\frac{9}{22}$	$\frac{1}{22}$

$$\begin{aligned} 3. \quad F(x) &= 0, & -\infty < x < 0 \\ &= 1/8, & 0 \leq x < 1 \\ &= 1/2, & 1 \leq x < 2 \\ &= 7/8, & 2 \leq x < 3 \\ &= 1, & x \geq 3 \end{aligned}$$

Hint: See worked out Example 27.

$$4. \quad (i) 1/81 \quad (ii) 1/9, 8/9, 7/27 \quad (iii) 5 \quad 5. (i) 9/16, 7/16 \quad (ii) 0.45.$$

$$6. \quad 4/9. \quad 7. a = 1/8, \text{ Mean} = 3. \quad 9. 20, 2\sqrt{5}.$$

$$10. \quad y_0 = 3/4, \text{ Mean} = 1, \text{ Variance} = 1/5.$$

$$11. (i) 1/2 \quad (ii) 0, 2. \quad 12. 1/3, 2/9.$$

13.	$X = x$	:	0	1	2	3	4
	$P(X = x)$	:	$\frac{1}{42}$	$\frac{5}{21}$	$\frac{10}{21}$	$\frac{5}{21}$	$\frac{1}{42}$

$$14. \quad C = \frac{1}{\sqrt{2\pi}}, \text{ Expectation} = 2, \text{ Variance} = 1.$$